

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-II)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B	A,D	A,B,C	A,B,D	A,B	A,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	A	B	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	7.66 to 7.67	4.60 to 4.70	3.90 to 4.10	2.95 to 3.05	7.06 to 7.12	2.12 to 2.16	1.98 to 2.02	1.03 to 1.07

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,C or C	B,D	B,C	B,C,D	B,C	A,B		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	A	C	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	92.00 to 93.00	0.47 to 0.48	10.15	0.22 to 0.23	4.77	1.78	3.59 or 3.60	68.00

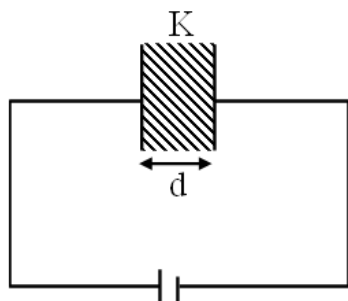
PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,C	B,D	B,D	A,C	A,C	A,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	D	A	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	3.14	1.50	0.79 to 0.81	2.50	569.00	2.50	2.00	3.00

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

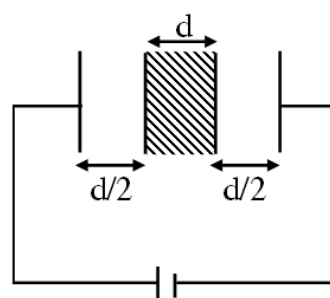
1. Ans (B)

For figure(a)



$$E_0 = \frac{V}{d}; C = \frac{K\epsilon_0 A}{d}$$

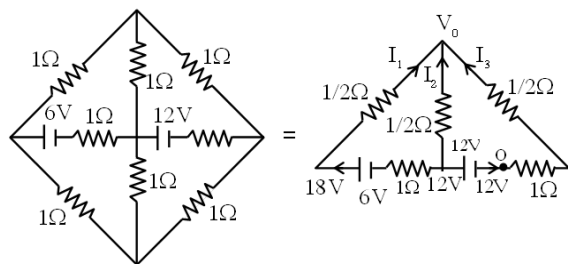
For figure(b)



$$C' = \frac{\epsilon_0 A}{2d - d + d/k};$$

$$C' = \frac{K\epsilon_0 A}{(K+1)d}; C' = \frac{C}{K+1}$$

2. Ans (A,D)



From KCL

$$i_1 + i_2 + i_3 = 0$$

$$\Rightarrow \frac{18 - V_0}{3/2} + \frac{12 - V_0}{1/2} + \frac{0 - V_0}{3/2} = 0$$

$$\Rightarrow 18 - V_0 + 36 - 3V_0 - V_0 = 0$$

$$\Rightarrow 54 = 5V_0$$

$$\frac{2\left(\frac{54}{5} - v'\right)}{1} + \frac{18 - v'}{1} = 0$$

$$\Rightarrow \frac{108}{5} + 18 = 3v'$$

$$\Rightarrow v' = \frac{198}{5 \times 3} = \frac{66}{5} \text{ V}$$

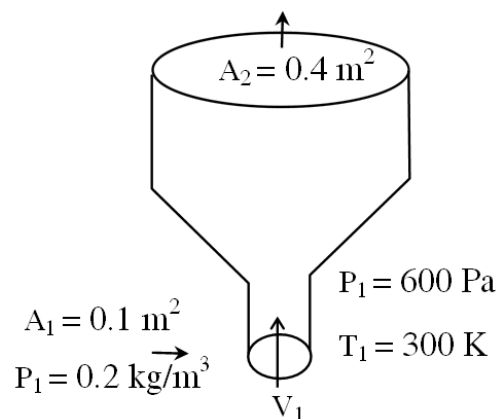
$$I_{R_1} = \frac{36}{5} = 7.2 \text{ A}$$

$$I_{R_2} = \frac{6}{5} = 1.2 \text{ A}$$

$$I_{R_3} = \frac{24}{5} = 4.8 \text{ A}$$

$$I_{R_5} = \frac{12}{5} = 2.4 \text{ A}$$

3. Ans (A,B,C)



$$\frac{dm}{dt} = \rho_1 A_1 v_1 = 0.8 \text{ kg/s}$$

$$v_1 = \frac{0.8}{0.2 \times 0.1} = 40 \text{ m/s}$$

$$g = 10 \text{ m/s}^2$$

$$\gamma = 2$$

Gas undergoes adiabatic expansion,

$$p^{1-\gamma} T^\gamma = \text{Constant}$$

$$\frac{P_2}{P_1} = \left(\frac{T_1}{T_2}\right)^{\frac{\gamma}{1-\gamma}}$$

$$P_2 = \left(\frac{300}{150}\right)^{\frac{2}{-1}} \times 600$$

$$P_2 = \frac{600}{4} = 150 \text{ Pa}$$

$$\text{Now } \rho = \frac{PM}{RT} \Rightarrow \rho \propto \frac{P}{T}$$

$$\frac{\rho_1}{\rho_2} = \left(\frac{P_1}{P_2}\right) \left(\frac{T_1}{T_2}\right) = \left(\frac{150}{600}\right) \left(\frac{300}{150}\right) = \frac{1}{2}$$

$$\rho_2 = \frac{\rho_1}{2} = 0.1 \text{ kg/m}^3$$

$$\text{Now } \rho_2 A_2 v_2 = 0.8 \Rightarrow v_2 = \frac{0.8}{0.1 \times 0.4} = 20 \text{ m/s}$$

$$\text{Now } W_{\text{on gas}} = \Delta K + \Delta U + (\text{Internal energy})$$

$$P_1 A_1 \Delta x_1 - P_2 A_2 \Delta x_2 = \frac{1}{2} \Delta m v_2^2 - \frac{1}{2} \Delta m v_1^2 + \Delta m g h$$

$$+ \frac{f}{2} (P_2 \Delta V_2 - P_1 \Delta V_1)$$

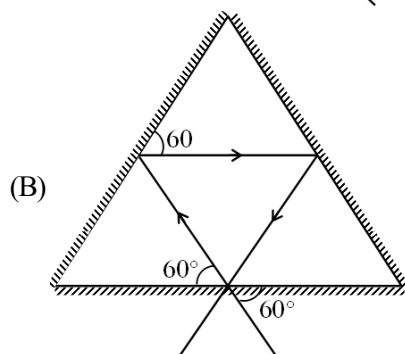
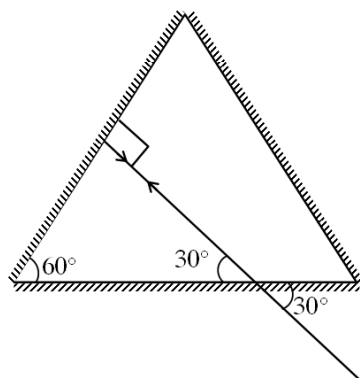
$$\Rightarrow 2P_1 \frac{\Delta V_1}{\Delta m} - 2P_2 \frac{\Delta V_2}{\Delta m} = \frac{v_2^2 - v_1^2}{2} + gh$$

$$\Rightarrow \frac{2 \times 600}{0.2} - \frac{2 \times 150}{0.1} = \frac{20^2 - 40^2}{2} + 10h$$

$$h = 360 \text{ m}$$

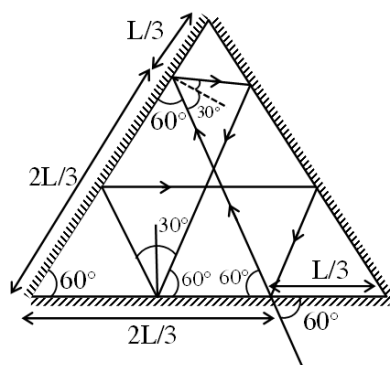
4. Ans (A,B,D)

(A) Ray will come out after one reflection for $\theta = 30^\circ$
& $0 < \ell < L$

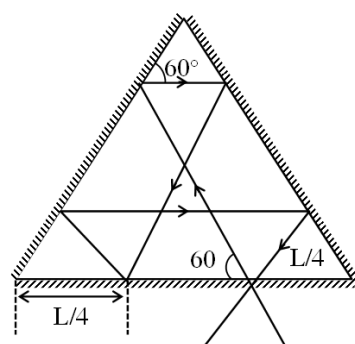


for $\theta = 60^\circ$ & $\ell = \frac{L}{2}$, ray will come out after two reflections.

(C) For $\ell = \frac{L}{3}$ & $\theta = 60^\circ$ ray will come out after five reflections.

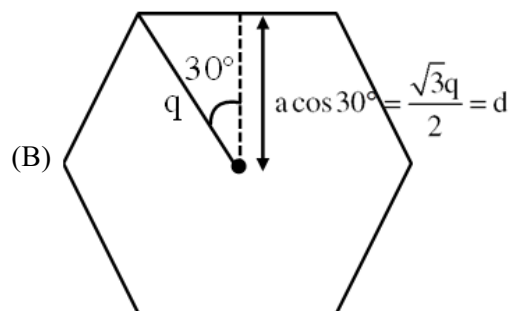


(D) For $\theta = 60^\circ$ & $0 < \ell < \frac{L}{2}$, ray will come out after five reflections



5. Ans (A,B)

(A) Due to symmetry $\vec{E}_0 = 0$



$$E_{\text{net}} = \frac{kq}{(2d)^2} \times 2 = \frac{2q \times 4}{4\pi\epsilon_0 \cdot 4 \cdot 3a^2}$$

$$= \frac{q}{6\pi\epsilon_0 a^2}$$

$$(C) v = \frac{7kq}{2d} = \frac{7q}{4\pi\epsilon_0 \cdot \sqrt{3}a} = \frac{7q}{4\sqrt{3}\pi\epsilon_0 q}$$

$$(D) v = \frac{2kq}{2d} = \frac{2q}{4\pi\epsilon_0 \cdot \sqrt{3}a} = \frac{q}{2\sqrt{3}\pi\epsilon_0 q}$$

6. Ans (A,D)

Binding energy of proton & neutron due to nuclear force is same. So difference in binding energy is only due to electrostatic P.E. and it is positive

$$E_0^p - E_0^n = \text{electrostatic P.E.}$$

$$= Z \times \text{P.E. of one proton}$$

$$= Z \times \frac{1}{4\pi\epsilon_0} \frac{(Z-1)e^2}{R}$$

$$\text{Where } R = R_0 A^{1/3}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{Z(Z-1)e^2}{R_0 A^{1/3}}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (A)

$$(I) \vec{B} = \mu_0 m I (\sin \omega t \hat{j} + \cos \omega t \hat{k})$$

$$\phi = \vec{B} \cdot \vec{A} = \mu_0 m I \cos(\omega t) \cdot A$$

$$\varepsilon = \frac{d\phi}{dt} = \mu_0 m I \omega A \sin(\omega t)$$

$$i = \frac{\varepsilon}{R} = \frac{\mu_0 m I \omega A}{R} \sin(\omega t)$$

$$\vec{M} = i \vec{A} = i A (\hat{k}) = \frac{\mu_0 m I \omega A^2}{R} \sin(\omega t) (\hat{k})$$

$$\vec{\tau} = \vec{M} \times \vec{B} = \frac{\mu_0 m^2 I^2 \omega A^2}{R} \sin^2(\omega t) (-\hat{i})$$

$$= -\left(\frac{\alpha}{4}\right) \hat{i}$$

$$(II) \vec{B} = \mu_0 m I (\sin \omega t \hat{i} + \cos \omega t \hat{j})$$

$$\phi = 0, \varepsilon = 0, i = 0, t = 0$$

$$(III) \vec{B} = \mu_0 m I (\sin \omega t \hat{i} + \cos \omega t \hat{k})$$

$$\phi = \vec{B} \cdot \vec{A} = \mu_0 m I \cdot \cos(\omega t) \cdot A$$

$$\varepsilon = -\frac{d\phi}{dt} = \mu_0 m I \omega A \sin(\omega t)$$

$$i = \frac{\varepsilon}{R} = \frac{\mu_0 m I \omega A}{R} \sin(\omega t)$$

$$\vec{M} = i \vec{A} = i A (\hat{k}) = \frac{\mu_0 m I \omega A^2}{R} \sin(\omega t) (\hat{k})$$

$$\vec{\tau} = \vec{M} \times \vec{B} = \frac{\mu_0 m^2 I^2 \omega A^2}{R} \sin^2(\omega t) (+\hat{j})$$

$$= \frac{\alpha}{4} \hat{j}$$

$$(IV) \vec{B} = \mu_0 m I (\cos \omega t \hat{i} + \sin \omega t \hat{k})$$

$$\phi = \vec{B} \cdot \vec{A} = \mu_0 m I \cdot \sin(\omega t) \cdot A$$

$$\varepsilon = -\frac{d\phi}{dt} = \mu_0 m I \omega A \cos(\omega t)$$

$$i = \frac{\varepsilon}{R} = -\frac{\mu_0 m I \omega A}{R} \cos(\omega t)$$

$$\vec{M} = i \vec{A} = i A (\hat{k}) = -\frac{\mu_0 m I \omega A^2}{R} \cos(\omega t) (\hat{k})$$

$$\vec{\tau} = \vec{M} \times \vec{B} = -\frac{\mu_0 m^2 I^2 \omega A^2}{2R} \cos^2(\omega t) (\hat{j})$$

$$= \alpha \cdot \cos^2\left(\frac{\pi}{6}\right) \hat{i} = \frac{-3\alpha}{4} \hat{j}$$

8. Ans (A)

$$(I) v_{BA}^2 = v_A^2 + v_B^2 - 2v_{AB} \cos \theta$$

$$\text{As } \omega_A = \omega_B, \theta = 90^\circ \text{ remains constant.}$$

$$\text{Also, } v_A = v_B = \sqrt{5} \text{ m/s}$$

$$\text{So, } v_{BA} = \sqrt{10} \text{ m/s}$$

$$(II) \vec{u}_A = \frac{5\pi}{2} \hat{i} + \frac{5\pi}{2} \hat{j}$$

$$\vec{v}_A = \frac{5\pi}{2} \hat{i} + \left(\frac{5\pi}{2} - 10 \cdot \frac{\pi}{3}\right) \hat{j}$$

$$= \frac{5\pi}{2} \hat{i} - \frac{5\pi}{6} \hat{j}$$

$$\vec{u}_B = -\frac{5\pi}{2} \hat{i} + \frac{5\pi}{2} \hat{j}$$

$$\vec{u}_B = -\frac{5\pi}{2} \hat{i} - \left(\frac{5\pi}{6} + 1\right) \hat{j}$$

$$\vec{v}_{B,A} = -5\pi \hat{i} - \hat{j}$$

$$v_{BA} = \sqrt{25\pi^2 + 1}$$

$$(III) x_A = \sin t$$

$$v_A = \cos t = \frac{1}{2} \text{ m/s}$$

$$x_B = \cos t$$

$$v_B = -\sin t = -\frac{\sqrt{3}}{2} \text{ m/s}$$

$$v_{BA} = -\frac{\sqrt{3}}{2} - \frac{1}{2}$$

$$(IV) \vec{v}_A \& \vec{v}_B \text{ are always perpendicular}$$

$$\text{So, } |\vec{v}_{BA}| = \sqrt{v_A^2 + v_B^2} = \sqrt{2} \text{ m/s}$$

9. Ans (B)

$$(I) \Delta U = \Delta Q - \Delta W$$

$$= \left\{ (2 \times 10^{-3} \times 2250) - \frac{10^5 (10^{-3} - 10^{-6})}{10^3} \right\} \text{ kJ}$$

$$= 4.30 \text{ kJ}$$

$$(II) \Delta U = nC_V \Delta T = \frac{5}{2} nR \Delta T$$

$$= \frac{5}{2} \cdot (0.1) (8) (1500 - 500) \text{ J} = 2 \text{ kJ}$$

$$(III) P_1 V_2^\gamma = P_2 V_2^\gamma$$

$$\Rightarrow 2 \left(\frac{1}{3} \right)^{5/3} = P_2 \left(\frac{1}{24} \right)^{5/3}$$

$$\Rightarrow P_2 = 64 \text{ kPa}$$

$$\Delta U = nC_V \Delta T = \frac{3}{2} \cdot (P_2 V_2 - P_1 V_1)$$

$$= \frac{3}{2} \left(64 \times \frac{1}{24} - 2 \times \frac{1}{3} \right) \text{ kJ} = 3 \text{ kJ}$$

$$(IV) \Delta U = nC_V \Delta T$$

$$= n \cdot \frac{7}{2} R \Delta T = \frac{7}{9} \Delta Q = 7 \text{ kJ}$$

10. Ans (A)

$$(I) v_1 = \frac{uf}{u+f}$$

$$= \frac{(-20)(10)}{(-20) + (10)} = +20$$

$$u_2 = +15$$

$$v_2 = \frac{(15)(15)}{(15) + (15)} = +7.5$$

$$(II) v_1 = +20$$

$$u_2 = +15$$

$$v_2 = \frac{(15)(-10)}{(15) + (-10)} = -30$$

$$(III) v_1 = +20$$

$$u_2 = +15$$

$$v_2 = \frac{(15)(-20)}{(15) + (-20)} = 60$$

$$(IV) v_1 = \frac{(-20)(-20)}{(-20) + (-20)} = -10$$

$$u_2 = -15$$

$$v_2 = \frac{(-15)(10)}{(-15) + (10)} = 30$$

PART-1 : PHYSICS

SECTION-II

1. Ans (7.66 to 7.67)

$$\text{Given } R_A = R_B = R$$

$$M_B = 2M_A$$

Calculation of escape velocity for A:

$$\text{Radius of remaining star} = \frac{R_A}{2}.$$

$$\text{Mass of remaining star} = \rho_A \frac{4}{3} \pi \frac{R_A^3}{8} = \frac{M_A}{8}$$

$$\frac{-GM_{A/B}}{R_{A/2}} + \frac{1}{2} mv_A^2 = 0$$

$$\Rightarrow v_A = \sqrt{\frac{2GM_{A/8}}{R_{A/2}}} = \sqrt{\frac{GM_A}{2R}}$$

Calculation of escape velocity for B

$$\text{Mass collected over B} = \frac{7}{8} M_A$$

Let the radius of B becomes r.

$$\therefore \frac{4}{3} \pi (r^3 - R_B^3) \rho_A = \frac{7}{8} \rho_A \frac{4}{3} \pi R_A^3$$

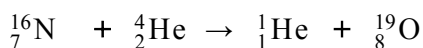
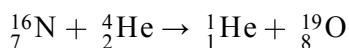
$$\Rightarrow r^3 = \frac{7}{8} R_A^3 + R_B^3 = \frac{27}{8} R_A^3$$

$$\Rightarrow r = \frac{3}{2} R$$

$$V_B = \sqrt{\frac{2GM}{r}} = \sqrt{\frac{2G \cdot 23M_A}{8 \cdot \frac{3R}{2}}} = \sqrt{\frac{23M_A}{6R}}$$

$$\frac{V_B}{V_A} = \sqrt{\frac{23}{3}}$$

2. Ans (4.60 to 4.70)



$$16.004 \quad 4.003 \quad 1.008 \quad 19.003$$

$$4v_0 = 1v_1 + 19v_2 = 20v_2 \quad (\text{For max loss of KE})$$

$$v_0 = \frac{v_2}{5}$$

$$E_{\text{required}} = (1.008 + 19.003 - 16.004 - 4.003) \times 930 = 3.72$$

$$\frac{1}{2}4v_0^2 - \frac{1}{2}20v_2^2 = 3.72$$

$$\frac{1}{2}4v_0^2 - 10\frac{v_0^2}{25}20v_2^2 = 3.72$$

$$2v_0^2 - \frac{2}{5}v_0^2 = 3.72$$

$$\frac{8}{5}v_0^2 = 3.72$$

$$v_0^2 = \frac{3.72 \times 5}{8}$$

$$\text{KE} = \frac{1}{2}4v_0^2 = 2v_0^2 = \frac{3.72 \times 5}{4}$$

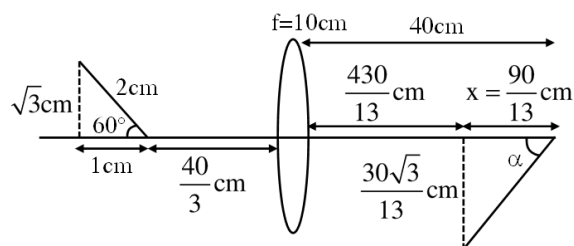
$$= 4.65$$

3. Ans (3.90 to 4.10)

Potential difference across the terminals of C_3 is 2V.

$$\therefore Q_3 = CV = (2\mu)(2) = 4\mu\text{C}$$

4. Ans (2.95 to 3.05)



$$\frac{h_i}{h_o} = \frac{v}{u} \Rightarrow \frac{-\frac{30\sqrt{3}}{13}}{\sqrt{3}} = \frac{v}{-\frac{43}{3}} \Rightarrow v_1 = \frac{430}{13} \text{ cm}$$

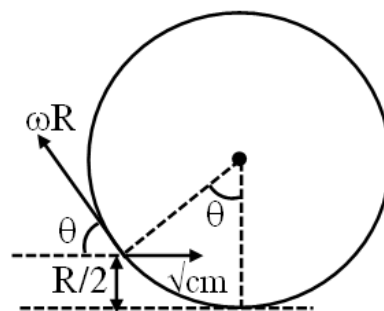
$$* \frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} = \frac{1}{10} - \frac{3}{40} \Rightarrow v = 40 \text{ cm}$$

$$* x = 40 - \frac{430}{13} = \frac{90}{13} \text{ cm}$$

$$\tan \alpha = \frac{\frac{30\sqrt{3}}{13}}{\frac{90}{13}} = \frac{1}{\sqrt{3}} \Rightarrow \alpha = 30^\circ = \frac{\pi}{6}$$

N = 3 Ans.

5. Ans (7.06 to 7.12)



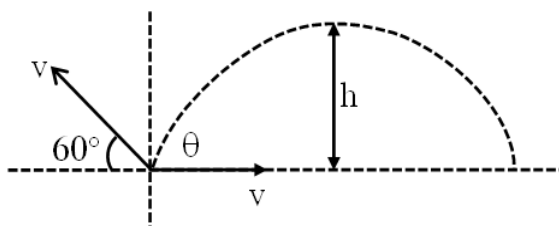
$$\text{At } t = 0, \omega = 0$$

$$\text{at } t = \sqrt{\pi}, \omega = at = \frac{2}{3}\sqrt{\pi}, v = \omega r = \frac{4}{3}\sqrt{\pi}$$

$$\theta = \frac{1}{2}at^2$$

$$\theta = \frac{1}{2} \times \frac{2}{3} \times \pi = \frac{\pi}{3}$$

$$\theta = 60^\circ$$



$$v_y = v \sin 60 = \frac{\sqrt{3}}{2}v$$

$$h = \frac{u_y^2}{2g} = \frac{\frac{3}{4}v^2}{2g}$$

$$h = \frac{\frac{3}{4} \times \frac{16}{9}\pi}{2g}$$

$$h = \frac{3\pi}{9 \times 2g} = \frac{4\pi}{6g}$$

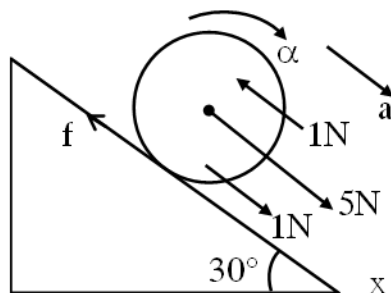
$$\text{Maximum height from plane, } H = \frac{R}{2} + h$$

$$H = 1 + \frac{\pi}{15}$$

$$x = 7.09$$

6. Ans (2.12 to 2.16)

Solid sphere 1kg, 1m



$$5 + 1 - 1 - f = 1a$$

$$5 - f = a$$

About COM

$$f \cdot 1 - 2(1(0.5)) = \frac{2}{5} Mr^2 \alpha$$

$$\Rightarrow f - 1 = \frac{2}{5} a \Rightarrow f = 1 + \frac{2}{5} a$$

$$5 - a = 1 + \frac{2}{5} a$$

$$\Rightarrow 4 = \frac{7a}{5} \Rightarrow a = \frac{20}{7}$$

$$f = 1 + \frac{8}{7} = 2.14$$

7. Ans (1.98 to 2.02)

Maximum energy will be

$$\frac{q_0^2}{2C} = \frac{1}{2} LI_0^2$$

$$\frac{q_0^2}{CL} = I_0^2$$

$$I_0 = \frac{q_0}{\sqrt{LC}}$$

$$I_0 = \frac{CV}{\sqrt{LC}}$$

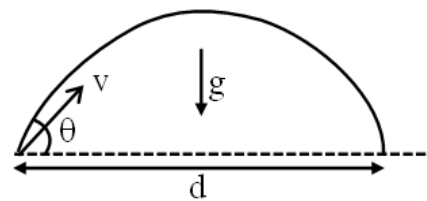
$$I_0 = \sqrt{\frac{C}{L}} \times V$$

$$V = \text{emf} = \left| \frac{AdB}{dt} \right|$$

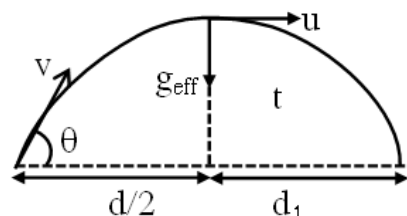
$$I_0 = \sqrt{\frac{10^{-3}}{0.4}} \times 0.04 \quad V = (0.002)$$

Maximum current $I_0 = 2\text{mA}$

8. Ans (1.03 to 1.07)



$$d = \frac{v^2 \sin 2\theta}{g}$$



$$H_{\max} = \frac{v^2 \sin^2 \theta}{2g} ; \frac{1}{2} g_{\text{eff}} t^2 = H_{\max}$$

$$\Rightarrow t^2 = \frac{2H_{\max}}{g_{\text{eff}}} ; t = \sqrt{\frac{v^2 \sin^2 \theta \times 0.81}{g^2}} ;$$

$$t = \frac{0.9v \sin \theta}{g}$$

$$t^2 = \frac{2 \times v^2 \sin^2 \theta}{2g \left(\frac{g}{0.81} \right)}$$

$$d' = \text{New range} = \frac{d}{2} + d_1$$

$$d_1 = v \cos \theta t$$

$$= \frac{v^2 \sin^2 \theta \cos \theta \times 0.9}{g} ;$$

$$d' = \frac{v^2 \sin 2\theta}{2g} + \frac{v^2 \sin 2\theta \times 0.9}{2g}$$

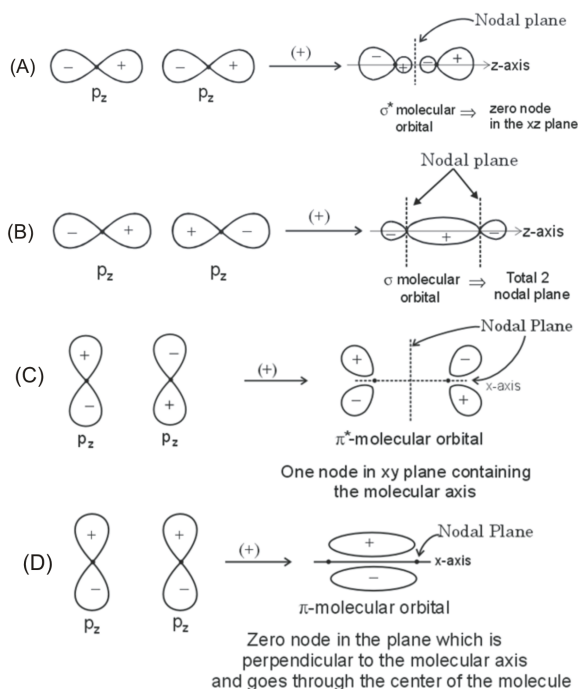
$$= \frac{v^2 \sin 2\theta}{g} \left(\frac{1.0}{2} \right) = 0.95d$$

$$n = 1.05$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (B,C or C)

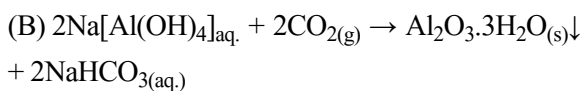


2. Ans (B,D)

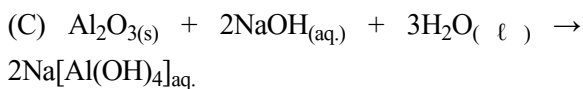
- (A) Chemisorption is unimolecular layered.
(B) Enthalpy of physisorption is much less in magnitude.
(C) Chemisorption of gases on solids is exothermic.
(D) As physisorption is exothermic so lowering temperature favours it.

3. Ans (B,C)

(A) Electrochemical extraction of Aluminum from bauxite is not preferred by reaction with coke



The sodium aluminate present in solution is neutralised by passing CO_2 gas and hydrated Al_2O_3 is precipitated.



Concentration of bauxite is carried out by heating the powdered ore with hot concentrated solution of NaOH

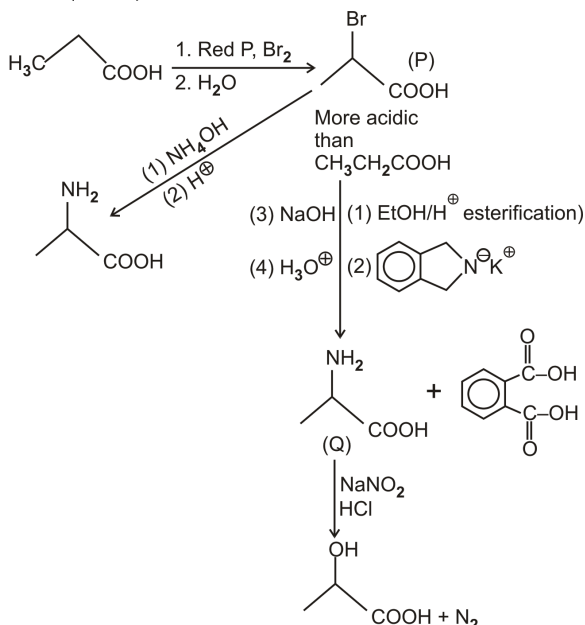
(D) In metallurgy of aluminum, Al_2O_3 is mixed with Na_3AlF_6

4. Ans (B,C,D)

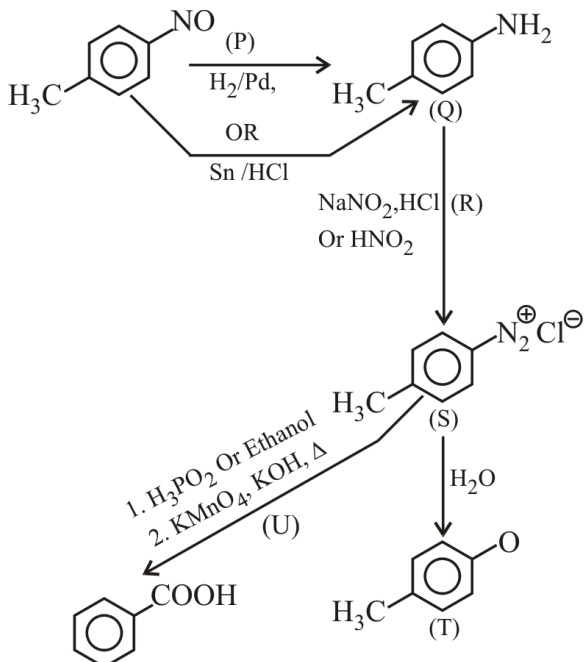


$\text{NO} \Rightarrow$ Neutral oxide, Paramagnetic, Linear geometry, Colourless gas

5. Ans (B,C)



6. Ans (A,B)



PART-2 : CHEMISTRY

SECTION-I (ii)

7. **Ans (A)**

$$(I) \text{ rate} = \frac{k[x]}{x_s + [x]} = \frac{k}{\frac{x_s}{[x]} + 1}$$

If $[x] \rightarrow \infty \Rightarrow \text{rate} \rightarrow k \Rightarrow \text{order} = 0$

$\Rightarrow (I) - (R), (P)$

$$(II) [x] \ll x_s \Rightarrow \text{rate} = \frac{k[x]}{x_s} \Rightarrow \text{order} = 1$$

$\Rightarrow (II) - (Q), (T)$

(III) $[x] > x_s \Rightarrow \text{rate} = k \Rightarrow \text{order} = 0$

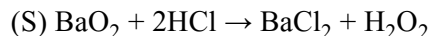
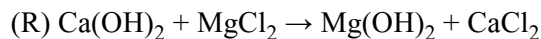
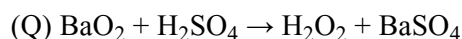
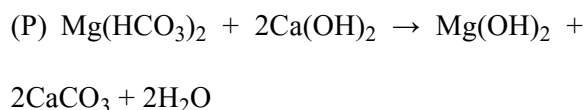
$\Rightarrow (III) - (P), (S)$

$$(IV) \text{ rate} = \frac{k[x]^2}{x_s + [x]}$$

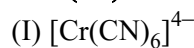
$[x] \gg x_s \Rightarrow \text{rate} = k[x]$

$\Rightarrow (IV) - (Q), (T)$

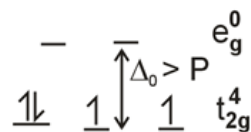
8. **Ans (A)**



9. **Ans (C)**



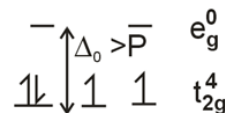
$\text{Cr}^{+2} = [\text{Ar}]_{18} 3d^4 4s^0$; low spin complex



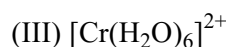
P, R, T



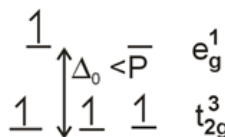
$\text{Ru}^{+4} = [\text{Kr}]_{36} 4d^4 5s^0$; low spin complex



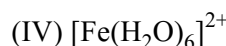
P, R, S, T



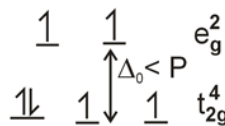
$\text{Cr}^{+2} = [\text{Ar}]_{18} 3d^4 4s^0$; high spin complex



Q, T

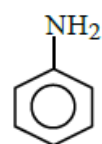


$\text{Fe}^{+2} = [\text{Ar}]_{18} 3d^6$; High spin complex



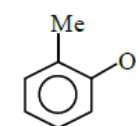
P, Q

10. **Ans (C)**



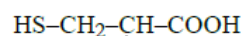
: Blue colour in Lassaign test due to presence of N

Aniline



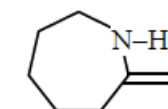
: Violet colour with FeCl_3 due to presence of phenolic OH

o-Cresol



NH_2
Cysteine

: It gives blood red colour with NaSCN + Sod. nitroprusside.



: Blue colour in Lassaign test

Caprolactam

due to presence of N

PART-2 : CHEMISTRY

SECTION-II

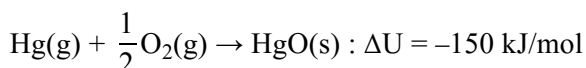
1. **Ans (92.00 to 93.00)**

$$Q_{\text{rxn}} = C\Delta T$$

$$|\Delta U| \times 2 = 20 \times 15$$

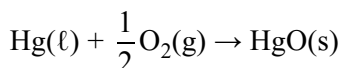
$$|\Delta U| = 150 \text{ kJ/mol}$$

$$\Delta U = -150 \text{ kJ/mol}$$



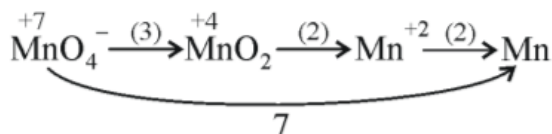
$$\Delta H = \Delta U + \Delta n_g RT$$

$$= -150 - \frac{3}{2} \times \frac{8}{1000} \times 298 = -153.57$$



$$\Delta H = -153.57 + 61.57 = -92.00 \text{ kJ/mol}$$

2. **Ans (0.47 to 0.48)**



$$\text{For the required reaction } \Delta G^\circ = \Delta G^\circ_1 + \Delta G^\circ_2 + \Delta G^\circ_3$$

$$\Rightarrow 7 \times E = 1.68 \times 3 + 1.21 \times 2 + (-1.03) \times 2$$

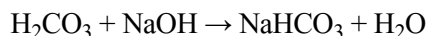
$$E = \frac{5.4}{7} = 0.7714$$

Similarly for "Y"

$$E = 1.77/6 = 0.295$$

$$\text{Ans.} = 0.48$$

3. **Ans (10.15)**



Milli moles	20	20	10
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At end	0	0	20 + 10 = 30
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Final mixture has 30 milli moles NaHCO_3 and 20 milli moles Na_2CO_3

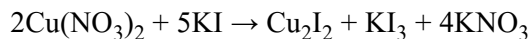
$$\text{pH} = \text{pK}_{a2} + \log \frac{\text{Salt}}{\text{Acid}}$$

$$\text{pH} = \text{pK}_{a2} + \log \left(\frac{20}{30} \right)$$

$$[\text{Buffer} : \text{Na}_2\text{CO}_3 + \text{NaHCO}_3]$$

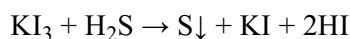
$$= 10.15$$

4. **Ans (0.22 to 0.23)**



$$0.02$$

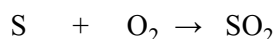
$$0.01$$



$$0.01$$

$$0.01$$

$$n_S = 0.01 \text{ mole}$$



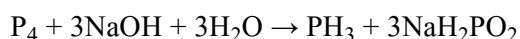
$$0.01$$

$$0.01$$

$$\text{volume at STP} = 0.01 \times 22.7 = 0.227 \text{ L}$$

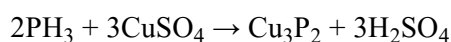
5. **Ans (4.77)**

$$\text{Mole of P}_4 = \frac{2.48}{31 \times 4} = 0.02$$



$$0.02 \text{ mole}$$

$$0.02 \text{ mole}$$



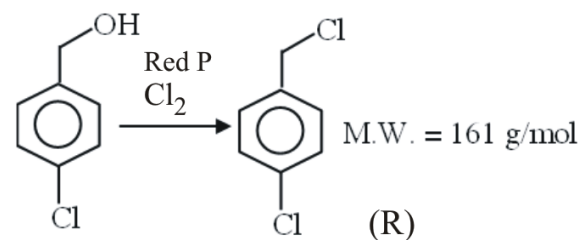
$$0.02 \quad \frac{3}{2} \times 0.02$$

$$= 0.03 \text{ moles}$$

$$W_{\text{CuSO}_4} = 0.03 \times 159 = 4.77$$

$$\text{Ans.} = 4.77$$

6. **Ans (1.78)**



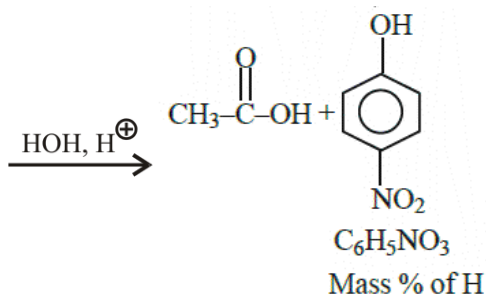
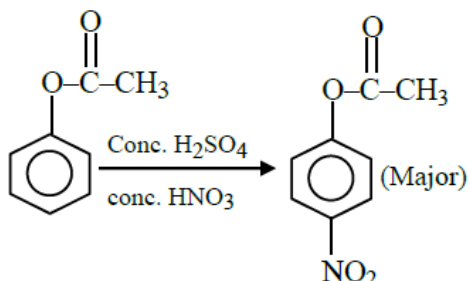
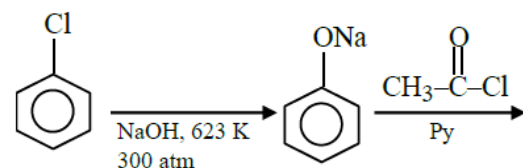
$$1 \text{ g R} \rightarrow \frac{1}{161} \text{ moles}$$

$$\text{No. of Cl Atoms} \rightarrow \frac{2}{161} \text{ moles}$$

$$\text{Moles of AgCl} \rightarrow \frac{2}{161} \text{ moles}$$

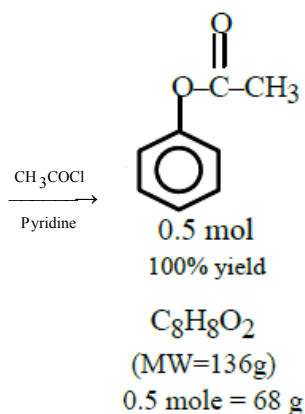
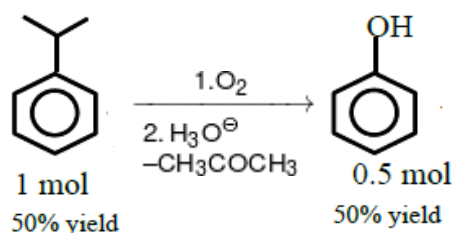
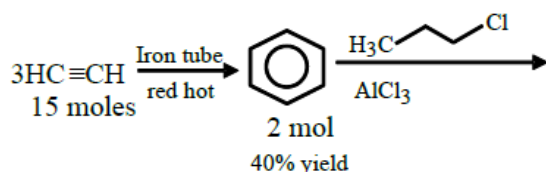
$$\text{Mass of AgCl} = \frac{2}{161} \times (108 + 35.5) = 1.78$$

7. Ans (3.59 or 3.60)



$$= \frac{5}{139} \times 100 = 3.597$$

8. Ans (68.00)



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,C)

$$\vec{r} = \hat{k} + t(-\hat{i} + \hat{j}) + p(-\hat{i} + \hat{k})$$

$$\vec{n} = \hat{i} + \hat{j} + \hat{k}$$

$$\Rightarrow x + y + z = 1$$

$$Q(10, 15, 20) \text{ and } S(\alpha, \beta, \gamma)$$

$$\frac{\alpha - 10}{1} = \frac{\beta - 15}{1} = \frac{\gamma - 20}{1} = -2 \left(\frac{10 + 15 + 20 - 1}{1 + 1 + 1} \right)$$

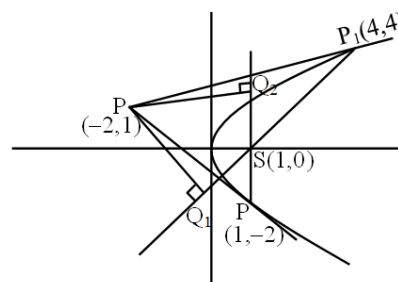
$$= -\frac{88}{3}$$

$$\Rightarrow (\alpha, \beta, \gamma) = \left(-\frac{58}{3}, -\frac{43}{3}, -\frac{28}{3} \right)$$

$\Rightarrow$ A, B, C are correct options

2. Ans (B,D)

Let equation of tangent with slope 'm' be



$$T : y = mx + \frac{1}{m}$$

T : passes through (-2, 1) so

$$1 = -2m + \frac{1}{m}$$

$$\Rightarrow m = -1 \text{ or } m = \frac{1}{2}$$

$$\text{Points are given by } \left(\frac{a}{m^2}, \frac{2a}{m} \right)$$

So, one point will be (1, -2) & (4, 4)

Let $P_1(4, 4)$ & $P_2(1, -2)$

$$P_1S : 4x - 3y - 4 = 0$$

$$P_2S : x - 1 = 0$$

$$PQ_1 = \left| \frac{4(-2) - 3(1) - 4}{5} \right| = 3$$

$$SP = \sqrt{10} ; PQ_2 = 3 ; SQ_1 = 1 = SQ_2$$

$$\frac{1}{2} \left(\frac{Q_1Q_2}{2} \right) \times \sqrt{10} = \frac{1}{2} \times 3 \times 1 \text{ (comparing Areas)}$$

$$\Rightarrow Q_1Q_2 = \frac{2 \times 3}{\sqrt{10}} = \frac{3\sqrt{10}}{5}$$

3. Ans (B,D)

$$f(\theta) = \frac{1}{2} \begin{vmatrix} 1 & \sin \theta & 1 \\ -\sin \theta & 1 & \sin \theta \\ -1 & -\sin \theta & 1 \end{vmatrix} + \begin{vmatrix} \sin \pi & \cos(\theta + \frac{\pi}{4}) & \tan(\theta - \frac{\pi}{4}) \\ \sin(\theta - \frac{\pi}{4}) & -\cos \frac{\pi}{2} & \log_e(\frac{4}{\pi}) \\ \cot(\theta + \frac{\pi}{4}) & \log_e \frac{\pi}{4} & \tan \pi \end{vmatrix}$$

$$f(\theta) = \frac{1}{2} \begin{vmatrix} 2 & \sin \theta & 1 \\ 0 & 1 & \sin \theta \\ 0 & -\sin \theta & 1 \end{vmatrix} + \begin{vmatrix} 0 & -\sin(\theta - \frac{\pi}{4}) & \tan(\theta - \frac{\pi}{4}) \\ \sin(\theta - \frac{\pi}{4}) & 0 & \log_e(\frac{4}{\pi}) \\ -\tan(\theta - \frac{\pi}{4}) & -\log_e(\frac{4}{\pi}) & 0 \end{vmatrix}$$

$$f(\theta) = (1 + \sin^2 \theta) + 0 \text{ (skew symmetric)}$$

$$g(\theta) = \sqrt{f(\theta) - 1} + \sqrt{f\left(\frac{\pi}{2} - \theta\right) - 1}$$

$$= |\sin \theta| + |\cos \theta| \quad \text{for } \theta \in \left[0, \frac{\pi}{2}\right]$$

$$g(\theta) \in [1, \sqrt{2}]$$

$$\text{Again let } P(x) = k(x - \sqrt{2})(x - 1)$$

$$2 - \sqrt{2} = k(2 - \sqrt{2})(2 - 1)$$

$$\Rightarrow k = 1 \quad (P(2) = 2 - \sqrt{2} \text{ given})$$

$$\therefore P(x) = (x - \sqrt{2})(x - 1)$$

$$\text{option (A) } P\left(\frac{5 - \sqrt{2}}{4}\right) > 0 \text{ incorrect}$$

$$\text{option (B) } P\left(\frac{5\sqrt{2} - 1}{4}\right) > 0 \text{ correct}$$

$$\text{option (C) } P\left(\frac{1 + 3\sqrt{2}}{4}\right) < 0 \text{ incorrect}$$

$$\text{for option (D) } P\left(\frac{3 + \sqrt{2}}{4}\right) < 0 \text{ correct}$$

4. Ans (A,C)

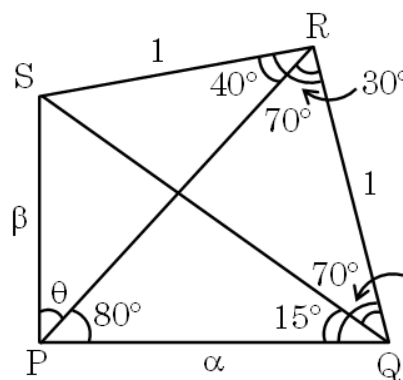
$$\angle PRQ = 70^\circ - 40^\circ = 30^\circ$$

$$\angle RQS = 70^\circ - 15^\circ = 55^\circ$$

$$\angle QSR = 180^\circ - 55^\circ - 70^\circ = 55^\circ$$

$$\therefore QR = RS = 1$$

$$\angle QPR = 180^\circ - 70^\circ - 30^\circ = 80^\circ$$



Apply sine-rule in ΔPRQ :

$$\frac{\alpha}{\sin 30^\circ} = \frac{1}{\sin 80^\circ} \Rightarrow \alpha = \frac{1}{2 \sin 80^\circ} \quad \dots(1)$$

Apply sine-rule in ΔPRS

$$\frac{\beta}{\sin 40^\circ} = \frac{1}{\sin \theta} \Rightarrow \beta \sin \theta = \sin 40^\circ \quad \dots(2)$$

$$4\alpha\beta \sin \theta = \frac{4 \sin 40^\circ}{2 \sin 80^\circ} = \frac{4 \sin 40^\circ}{2(2 \sin 40^\circ \cos 40^\circ)}$$

$$= \sec 40^\circ$$

$$\text{Now } \sec 30^\circ < \sec 40^\circ < \sec 45^\circ$$

$$\Rightarrow \frac{2}{\sqrt{3}} < \sec 40^\circ < \sqrt{2}$$

5. Ans (A,C)

$$\alpha = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^6 + \dots$$

$$\alpha = \frac{\frac{1}{4}}{1 - \frac{1}{4}} = \frac{1}{3}$$

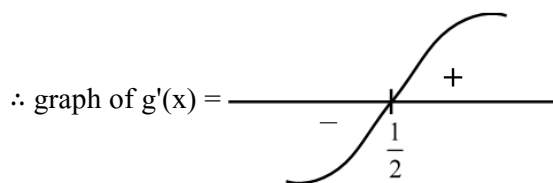
$$\therefore g(x) = 2^{x/3} + 2^{1/3(1-x)}$$

$$\therefore g(x) = 2^{x/3} + \frac{2^{1/3}}{2^{x/3}}$$

$$\text{where } g(0) = 1 + 2^{1/3} \text{ \& } g(1) = 1 + 2^{1/3}$$

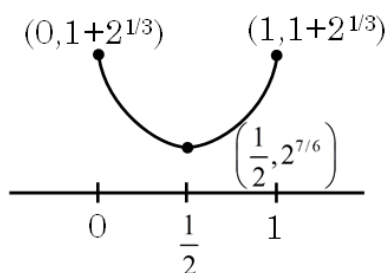
$$\therefore g'(x) = \frac{1}{3} \left(2^{x/3} - \frac{2^{1/3}}{2^{x/3}} \right) = 0$$

$$\Rightarrow 2^{2x/3} = 2^{1/3} \Rightarrow x = \frac{1}{2} = \text{critical point}$$



$$\& g\left(\frac{1}{2}\right) = 2^{7/6}$$

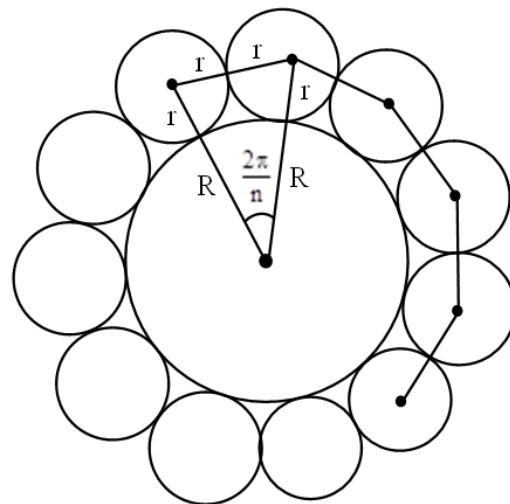
$\therefore$ graph of $g(x)$ in $[0, 1]$



6. Ans (A,C)

$$2(R+r) \sin \frac{\pi}{n} = 2r$$

$$\frac{R+r}{r} = \operatorname{cosec} \frac{\pi}{n}$$



(A) $n = 4, R + r = \sqrt{2}r$

(B) $n = 5, \frac{R+r}{r} = \operatorname{cosec} \frac{\pi}{5} < \operatorname{cosec} \frac{\pi}{6}$

$$R + r < 2r \Rightarrow r > R$$

(C) $n = 8, \frac{R+r}{r} = \operatorname{cosec} \frac{\pi}{8} > \operatorname{cosec} \frac{\pi}{4}$

$$R + r > \sqrt{2}r$$

(D) $n = 12, \frac{R+r}{r} = \operatorname{cosec} \frac{\pi}{12} = \sqrt{2}(\sqrt{3} + 1)$

$$R + r = \sqrt{2}(\sqrt{3} + 1)r$$

$$\sqrt{2}(\sqrt{3} + 1)r > R$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (C)

$$(I) \left\{ x \in \left[-\frac{2\pi}{3}, \frac{2\pi}{3} \right] : \cos x + \sin x = 1 \right\}$$

$$\cos x + \sin x = 1$$

$$\Rightarrow \frac{1}{\sqrt{2}} \cos x + \frac{1}{\sqrt{2}} \sin x = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \cos \left(x - \frac{\pi}{4} \right) = \cos \frac{\pi}{4}$$

$$\Rightarrow x - \frac{\pi}{4} = 2n\pi \pm \frac{\pi}{4}; n \in \mathbb{Z}$$

$$\Rightarrow x = 2n\pi; x = 2n\pi + \frac{\pi}{2}; n \in \mathbb{Z}$$

$$\Rightarrow x \in \left\{ 0, \frac{\pi}{2} \right\} \text{ in given range has two solutions}$$

$$(II) \left\{ x \in \left[-\frac{5\pi}{18}, \frac{5\pi}{18} \right] : \sqrt{3} \tan 3x = 1 \right\}$$

$$\sqrt{3} \tan 3x = 1 \Rightarrow \tan 3x = \frac{1}{\sqrt{3}} \Rightarrow$$

$$3x = n\pi + \frac{\pi}{6}$$

$$\Rightarrow x = (6n + 1) \frac{\pi}{18}; n \in \mathbb{Z}$$

$$\Rightarrow x \in \left\{ \frac{\pi}{18}, \frac{-5\pi}{18} \right\} \text{ in given range has two solutions}$$

$$(III) \left\{ x \in \left[-\frac{6\pi}{5}, \frac{6\pi}{5} \right] : 2 \cos(2x) = \sqrt{3} \right\}$$

$$2 \cos 2x = \sqrt{3}$$

$$\Rightarrow \cos 2x = \frac{\sqrt{3}}{2} = \cos \frac{\pi}{6}$$

$$\Rightarrow 2x = 2n\pi \pm \frac{\pi}{6}; n \in \mathbb{Z}$$

$$\Rightarrow x = n\pi \pm \frac{\pi}{12}; n \in \mathbb{Z}$$

$$x \in \left\{ \pm \frac{\pi}{12}, \pi \pm \frac{\pi}{12}, -\pi \pm \frac{\pi}{12} \right\}$$

Six solutions in given range

$$(IV) \left\{ x \in \left[-\frac{7\pi}{4}, \frac{7\pi}{4} \right] : \sin x - \cos x = 1 \right\}$$

$$\cos x - \sin x = -1$$

$$\Rightarrow \cos \left(x + \frac{\pi}{4} \right) = \frac{-1}{\sqrt{2}} = \cos \frac{3\pi}{4}$$

$$\Rightarrow x + \frac{\pi}{4} = 2n\pi \pm \frac{3\pi}{4}; n \in \mathbb{Z}$$

$$\Rightarrow x = 2n\pi + \frac{\pi}{2} \text{ or } x = 2n\pi - \pi; n \in \mathbb{Z}$$

$$\Rightarrow x \in \left\{ \frac{\pi}{2}, \frac{-3\pi}{2}, \pi, -\pi \right\} \text{ four solutions in given range}$$

8. Ans (D)

$$P(\text{draw in 1 round}) = \frac{6}{36} = \frac{1}{6}$$

$$P(\text{win in 1 round}) = \frac{1}{2} \left(1 - \frac{1}{6} \right) = \frac{5}{12}$$

$$P(\text{loss in 1 round}) = \frac{5}{12}$$

$$P(X_2 > Y_2) = P(10, 0) + P(7, 2)$$

$$= \frac{5}{12} \times \frac{5}{12} + \frac{5}{12} \times \frac{1}{6} \times 2 = \frac{45}{144} = \frac{5}{16}$$

$$P(X_2 = Y_2) = P(5, 5) + P(4, 4)$$

$$= \frac{5}{12} \times \frac{5}{12} \times 2 + \frac{1}{6} \times \frac{1}{6} = \frac{25 + 2}{72} = \frac{3}{8}$$

$$P(X_3 = Y_3) = P(6, 6) + P(7, 7)$$

$$= \frac{1}{6 \times 6 \times 6} + \frac{5}{12} \times \frac{1}{6} \times \frac{5}{12} \times 6 = \frac{2}{432} + \frac{75}{432}$$

$$= \frac{77}{432}$$

$$P(X_3 > Y_3) = \frac{1}{2} \left(1 - \frac{77}{432} \right) = \frac{355}{864}$$

9. Ans (A)

$$\text{If } \frac{q}{r} = 10 \Rightarrow D = 0 \Rightarrow D_x = D_y = D_z = 0$$

So, there are infinitely many solutions

Look of infinitely many solutions can be given as

$$x + y + z = 1 \quad \& \quad 10x + 100y + 1000z = 0$$

$$\Rightarrow x + 10y + 100z = 0$$

$$\text{Let } z = \lambda$$

$$\text{then } x + y = 1 - \lambda$$

$$\text{and } x + 10y = -100\lambda$$

$$\Rightarrow x = \frac{10}{9} + 10\lambda; y = \frac{-1}{9} - 11\lambda$$

$$\text{i.e., } (x, y, z) = \left(\frac{10}{9} + 10\lambda, \frac{-1}{9} - 11\lambda, \lambda \right)$$

$$Q \left(\frac{10}{9}, \frac{-1}{9}, 0 \right) \text{ valid for } \lambda = 0$$

$$P \left(0, \frac{10}{9}, \frac{-1}{9} \right) \text{ not valid for any } \lambda.$$

$$(I) \rightarrow Q, R, T \quad (II) \text{ If } \frac{p}{r} \neq 100, \text{ then } D_y \neq 0$$

So no solution

$$(II) \rightarrow (S)$$

$$(III) \text{ If } \frac{p}{q} \neq 10, \text{ then } D_z \neq 0 \text{ so, no solution}$$

$$(III) \rightarrow (S)$$

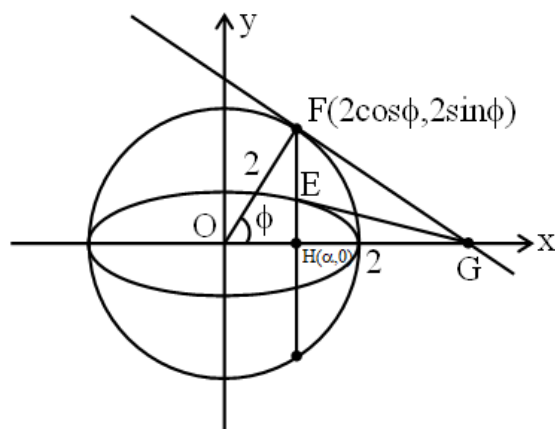
$$(IV) \text{ If } \frac{p}{q} = 10 \Rightarrow D_z = 0 \Rightarrow D_x = D_y = 0$$

so infinitely many solution

$$(IV) \rightarrow Q, R, T$$

10. Ans (A)

Let $F(2\cos\phi, 2\sin\phi)$

$$\& E(2\cos\phi, \quad \sqrt{3}\sin\phi)$$


$$\text{EG} : \frac{x}{2} \cos \phi + \frac{y}{\sqrt{3}} \sin \phi = 1$$

$$\therefore G\left(\frac{2}{\cos\phi}, 0\right) \text{ and } \alpha = 2\cos\phi$$

$$\text{ar}(\triangle FGH) = \frac{1}{2} \text{HG} \times \text{FH}$$

$$= \frac{1}{2} \left(\frac{2}{\cos \phi} - 2 \cos \phi \right) \times 2 \sin \phi$$

$$f(\phi) = 2 \tan \phi \sin^2 \phi$$

$$\therefore \text{(I) } f\left(\frac{\pi}{4}\right) = 1 \quad \text{(II) } f\left(\frac{\pi}{3}\right) = \frac{3\sqrt{3}}{2} \quad \text{(III)}$$

$$f\left(\frac{\pi}{6}\right) = \frac{1}{2\sqrt{3}}$$

$$\begin{aligned} \text{(IV) } f\left(\frac{\pi}{12}\right) &= 2(2 - \sqrt{3}) \left(\frac{\sqrt{3} - 1}{2\sqrt{2}}\right)^2 \\ &= (4 - 2\sqrt{3}) \frac{(\sqrt{3} - 1)^2}{8} = \frac{(\sqrt{3} - 1)^4}{8} \end{aligned}$$

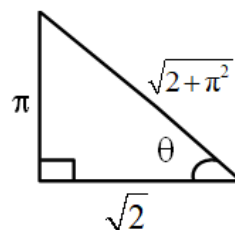
$$\therefore (I) \rightarrow (Q) ; (II) \rightarrow (T) ; (III) \rightarrow (S) ; (IV) \rightarrow (P)$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (3.14)**

$$\cos^{-1} \sqrt{\frac{2}{2+\pi^2}} = \tan^{-1} \frac{\pi}{\sqrt{2}}$$



$$\begin{aligned} \sin^{-1} \left(\frac{2\sqrt{2}\pi}{2+\pi^2} \right) &= \sin^{-1} \left(\frac{2 \times \frac{\pi}{\sqrt{2}}}{1 + \left(\frac{\pi}{\sqrt{2}} \right)^2} \right) \\ &= \pi - 2 \tan^{-1} \left(\frac{\pi}{\sqrt{2}} \right) \\ \left(\text{As, } \sin^{-1} \left(\frac{2x}{1+x^2} \right) &= \pi - 2 \tan^{-1} x, x \geq 1 \right) \end{aligned}$$

$$\begin{aligned} \text{and } \tan^{-1} \frac{\sqrt{2}}{\pi} &= \cot^{-1} \left(\frac{\pi}{\sqrt{2}} \right) \\ \therefore \text{Expression} &= \frac{3}{2} \left(\tan^{-1} \frac{\pi}{\sqrt{2}} \right) + \frac{1}{4} \left(\pi - 2 \tan^{-1} \frac{\pi}{\sqrt{2}} \right) + \cot^{-1} \left(\frac{\pi}{\sqrt{2}} \right) \\ &= \left(\frac{3}{2} - \frac{2}{4} \right) \tan^{-1} \frac{\pi}{\sqrt{2}} + \frac{\pi}{4} + \cot^{-1} \frac{\pi}{\sqrt{2}} \\ &= \left(\tan^{-1} \frac{\pi}{\sqrt{2}} + \cot^{-1} \frac{\pi}{\sqrt{2}} \right) + \frac{\pi}{4} \\ &= \frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4} \end{aligned}$$

2. Ans (1.50)

$$\lim_{x \rightarrow a^+} \frac{2 \ln(\sqrt{x} - \sqrt{a})}{\ln(e\sqrt{x} - e\sqrt{a})} \quad \left(\frac{0}{0} \text{form} \right)$$

$\therefore$ Using Lopital rule,

$$\begin{aligned} &= 2 \lim_{x \rightarrow \alpha^+} \frac{\left(\frac{1}{\sqrt{x}-\sqrt{\alpha}}\right) \cdot \frac{1}{2\sqrt{x}}}{\left(\frac{1}{e^{\sqrt{x}}-e^{\sqrt{\alpha}}}\right) \cdot e^{\sqrt{x}} \cdot \frac{1}{2\sqrt{x}}} \\ &= \frac{2}{e^{\sqrt{\alpha}}} \lim_{x \rightarrow \alpha^+} \frac{(e^{\sqrt{x}} - e^{\sqrt{\alpha}})}{(\sqrt{x} - \sqrt{\alpha})} \quad \left(\frac{0}{0}\right) \\ &= \frac{2}{e^{\sqrt{\alpha}}} \lim_{x \rightarrow \alpha^+} \frac{(e^{\sqrt{x}} \cdot \frac{1}{2\sqrt{x}} - 0)}{\left(\frac{1}{2\sqrt{x}} - 0\right)} = 2 \end{aligned}$$

$$\begin{aligned} \text{so, } \lim_{x \rightarrow \alpha^+} f(g(x)) &= \lim_{x \rightarrow \alpha^+} f(2) \\ &= f(2) = \sin \frac{\pi}{6} = \frac{1}{2} \\ &= 0.50 \end{aligned}$$

3. Ans (0.79 to 0.81)

$$n(U) = 900$$

Let A $\equiv$ Fever, B $\equiv$ Cough

C $\equiv$ Breathing problem

$$\therefore n(A) = 190, n(B) = 220, n(C) = 220$$

$$n(A \cup B) = 330, n(B \cup C) = 350,$$

$$n(A \cup C) = 340, n(A \cap B \cap C) = 30$$

$$\text{Now } n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$\Rightarrow 330 = 190 + 220 - n(A \cap B)$$

$$\Rightarrow n(A \cap B) = 80$$

Similarly,

$$350 = 220 + 220 - n(B \cap C)$$

$$\Rightarrow n(B \cap C) = 90$$

$$\text{and } 340 = 190 + 220 - n(A \cap C)$$

$$\Rightarrow n(A \cap C) = 70$$

$$\therefore n(A \cup B \cup C) = (190 + 220 + 220) - (80 + 90 + 70) + 30$$

$$= 660 - 240 = 420$$

$\Rightarrow$ Number of person without any symptom

$$= n(U) - n(A \cup B \cup C)$$

$$= 900 - 420 = 480$$

Now, number of person suffering from exactly one symptom

$$= (n(A) + n(B) + n(C)) - 2(n(A \cap B) + n(B \cap C) + n(C \cap A)) + 3n(A \cap B \cap C)$$

$$= (190 + 220 + 220) - 2(80 + 90 + 70) + 3(30)$$

$$= 630 - 480 + 90 = 240$$

$\therefore$ Number of person suffering from atmost one symptom

$$= 480 + 240 = 720$$

$\Rightarrow$ Probability

$$= \frac{720}{900} = \frac{8}{10} = \frac{4}{5} = 0.80$$

4. Ans (2.50)

Given that $z \neq \bar{z}$

$$\text{Let } \alpha = \frac{2 + 3z + 4z^2}{2 - 3z + 4z^2} = \frac{(2 - 3z + 4z^2) + 6z}{2 - 3z + 4z^2}$$

$$\therefore \alpha = 1 + \frac{6z}{2 - 3z + 4z^2}$$

If α is a real number, then

$$\alpha = \bar{\alpha}$$

$$\Rightarrow \frac{z}{2 - 3z + 4z^2} = \frac{\bar{z}}{2 - 3\bar{z} + 4\bar{z}^2}$$

$$\therefore 2(z - \bar{z}) = 4z\bar{z}(z - \bar{z})$$

$$\Rightarrow (z - \bar{z})(2 - 4z\bar{z}) = 0$$

As $z \neq \bar{z}$ (Given)

$$\Rightarrow z\bar{z} = \frac{2}{4} = \frac{1}{2}$$

$$\Rightarrow |z|^2 = 0.50$$

5. Ans (569.00)

$$(1) \begin{array}{|c|c|c|c|} \hline 2 & 0 & 2 & 2,3,4,6,7 \\ \hline \end{array} \rightarrow 5$$

$$(2) \begin{array}{|c|c|c|c|} \hline 2 & 0 & 3,4,6,7 & \\ \hline \end{array} \rightarrow 24$$

↓ ↓
4 6

$$(3) \begin{array}{|c|c|c|c|} \hline 2 & 2,3,4,6,7 & & \\ \hline \end{array} \rightarrow 180$$

↓ ↓ ↓
5 6 6

$$(4) \begin{array}{|c|c|c|c|} \hline 3 & & & \\ \hline \end{array} \rightarrow 216$$

↓ ↓ ↓
6 6 6

$$(5) \begin{array}{|c|c|c|c|} \hline 4 & 0,2,3,4 & & \\ \hline \end{array} \rightarrow 144$$

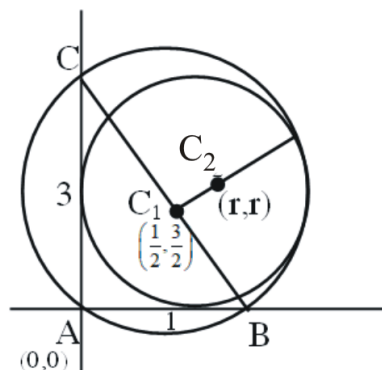
↓ ↓ ↓
4 6 6

Number of 4 digit integers in [2022,4482]

$$= 5 + 24 + 180 + 216 + 144 = 569$$

6. **Ans (2.50)**

$$4 - \sqrt{10} = 0.83 \text{ or } 0.84$$



$$C_1 \left(\frac{1}{2}, \frac{3}{2} \right) \text{ and } r_1 = \frac{\sqrt{10}}{2}$$

$$C_2 = (r, r)$$

$\therefore$ circle C_2 touches C_1 internally

$$\Rightarrow C_1 C_2 = \left| r - \frac{\sqrt{10}}{2} \right|$$

$$\Rightarrow \left(r - \frac{1}{2} \right)^2 + \left(r - \frac{3}{2} \right)^2 = \left(r - \frac{\sqrt{10}}{2} \right)^2$$

$$r^2 - 4r + \sqrt{10}r = 0$$

$$r = 0 \text{ (reject) or } r = 4 - \sqrt{10}$$

7. **Ans (2.00)**

$$x dy - (y^2 - 4y) dx = 0, x > 0$$

$$\int \frac{dy}{y^2 - 4y} = \int \frac{dx}{x}$$

$$\int \left(\frac{1}{y-4} - \frac{1}{y} \right) dy = 4 \int \frac{dx}{x}$$

$$\log_e |y-4| - \log_e |y| = 4 \log_e x + \log_e c$$

$$\frac{|y-4|}{|y|} = cx^4 \xrightarrow{(1,2)} c = 1$$

$$|y-4| = |y| x^4$$

$$C-1 \quad \text{and} \quad C-2$$

$$y-4 = yx^4$$

$$y-4 = -yx^4$$

$$y = \frac{4}{1-x^4}$$

$$y = \frac{4}{1+x^4}$$

$$y(1) = \text{ND (rejected)}$$

$$y(1) = 2$$

$$y(\sqrt{2}) = \frac{4}{5}$$

8. **Ans (3.00)**

$$\beta = \lim_{x \rightarrow 0} \frac{e^{x^3} - (1-x^3)^{1/3}}{\frac{x \sin^2 x}{x^2} x^2} + \frac{\left((1-x^2)^{1/2} - 1 \right) \sin x}{x \frac{\sin^2 x}{x^2} x^2}$$

use expansion

$$\beta = \lim_{x \rightarrow 0} \frac{(1+x^3) - \left(1 - \frac{x^3}{3}\right)}{x^3} + \lim_{x \rightarrow 0} \frac{\left(\left(1 - \frac{x^2}{2}\right) - 1 \right) \frac{\sin x}{x}}{x^2}$$

$$\beta = \lim_{x \rightarrow 0} \frac{4x^3}{3x^3} + \lim_{x \rightarrow 0} \frac{-x^2}{2x^2}$$

$$\beta = \frac{4}{3} - \frac{1}{2} = \frac{5}{6}$$

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-II)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	A	C	D				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	B	A,B,C	C,D	A,C,D	B	A,B,C,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	1	8	6	0	6	2	8	4

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	C	C	C	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,C,D	C,D	A,C	A,C,D	A,C,D	A,B,C		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	2	5	8	9	5	4	6	4

PART-3 : MATHEMATICS

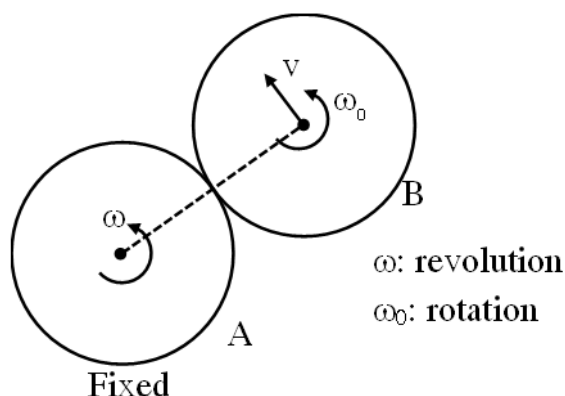
SECTION-I (i)	Q.	1	2	3	4				
	A.	D	B	D	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	D	B,D	A	C	A,B,C	A,B		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	4	9	1	4	8	7	4	9

HINT – SHEET**PART-1 : PHYSICS****SECTION-I (i)**1. **Ans (B)**

$$v = \omega (2R)$$

$$v = \omega_0 R : \text{no slipping}$$

$$\therefore \omega_0 = 2\omega$$



$$\vec{L} = m\vec{r} \times \vec{v}_c + I_c \omega_0$$

$$= M2Rv + MR^2 \omega_0$$

$$= 4MR^2 \omega + MR^2 (2\omega) = 6MR^2 \omega$$

$$\therefore n = 6$$

2. **Ans (A)**

$$\frac{hc}{\lambda} = \phi + 6 \quad \dots (i)$$

$$\frac{hc}{4\lambda} = \phi + 0.6 \quad \dots (ii)$$

$$\frac{3hc}{4\lambda} = 5.4\text{eV} \quad \therefore \phi = 1.2\text{ eV}$$

$$\Rightarrow \frac{3}{4} \times \frac{6.63 \times 10^{-24} \times 3 \times 10^8}{5.4 \times 1.6 \times 10^{-19}} = \lambda = 1.72 \times 10^{-7}\text{m}$$

3. **Ans (C)**

$$LC = \frac{0.1}{100} = 0.001 \text{ mm}$$

$$\text{Zero error} = 4 \times 0.001 = 0.004 \text{ mm}$$

$$\text{Reading 1} = 0.5 \times 4 + 20 \times 0.001 - 0.004 = 2.16 \text{ mm}$$

$$\text{Reading 2} = 0.5 \times 4 + 16 \times 0.001 - 0.004 = 2.12 \text{ mm}$$

$$\text{Mean value} = 2.14 \text{ mm}$$

$$\text{Mean absolute error} = \frac{0.02 + 0.02}{2} = 0.02$$

$$\text{Diameter} = 2.14 \pm 0.02$$

$$\text{Area} = \frac{\pi}{4} d^2$$

4. **Ans (D)**

$$\vec{B} = \frac{\mu_0 I}{4\pi L} (-\hat{k}) + \frac{\mu_0 I \pi}{4\pi \frac{L}{2}} (-\hat{k}) + \frac{\mu_0 I}{4\pi \frac{L}{4}} \times \frac{\pi}{2} (-\hat{k})$$

PART-1 : PHYSICS

SECTION-I (ii)

5. **Ans (B)**

$$q_1 = \int_0^1 k r 4\pi r^2 dr = \frac{4\pi k}{4} = \pi k$$

$$q_2 = \int_1^r \frac{2k}{r} 4\pi r^2 dr = \frac{8\pi k (r^2 - 1^2)}{2}$$

$$q_2 = 4\pi k [r^2 - 1] = 4\pi k r^2 - 4\pi k$$

$$q_{\text{net}} = q_1 + q_2$$

$$= 4\pi k r^2 - 3\pi k$$

$$q_{\text{net}} = \pi k [4r^2 - 3]$$

$$(A) E_{\text{net}} = 0 \Rightarrow q_{\text{net}} = 0 \Rightarrow r = \frac{\sqrt{3}}{2}$$

$$(B) V = \frac{kQ_{\text{net}}}{r} = \frac{1}{4\pi\epsilon_0} \frac{\pi k (4r^2 - 3)}{r}$$

$$V = \frac{k}{4\epsilon_0} \left[4r - \frac{3}{r} \right]$$

$$= \frac{k}{4\epsilon_0} \left[4 \times \frac{3}{2} - \frac{3 \times 2}{3} \right] = \frac{k}{\epsilon_0}$$

$$(C) q_{\text{net}} = \pi k [4(2)^2 - 3]$$

$$= 13\pi k$$

$$(D) E_2 = \frac{kQ}{r^2}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{\pi k (4r^2 - 3)}{r^2}$$

$$= \frac{k}{4\epsilon_0} \left[\frac{4\left(\frac{5}{2}\right)^2 - 3}{\left(\frac{5}{2}\right)^2} \right]$$

$$= \frac{k}{25\epsilon_0} [25 - 3] = \frac{22}{25} \frac{k}{\epsilon_0}$$

6. Ans (A,B,C)

Case (i)

When both switches are open equivalent resistance in circuit 1

$$R_{C_1} = \frac{16}{11} \Omega$$

Equivalent resistance in circuit 2

$$R_{C_2} = \frac{6}{11} \Omega$$

For voltage source

$$P = \frac{V^2}{R}$$

$$P \propto \frac{1}{R}$$

$$R_{C_1} > R_{C_2}$$

$$\Rightarrow P_2 > P_1 \text{ (Option (A) correct)}$$

For constant current source

$$P = i^2 R$$

$$P \propto R$$

$$\Rightarrow P_1 > P_2 \text{ (Option (B) correct)}$$

Case-II

When switch is closed

$$R'_{C_1} = \frac{5}{11} \Omega$$

$$R'_{C_2} = \frac{1}{2} \Omega$$

$$R'_{C_1} < R_{C_1}$$

For voltage source

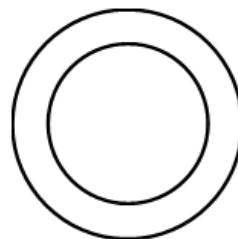
$$P \propto \frac{1}{R} \Rightarrow Q_1 > P_1 \text{ (Option (C) correct)}$$

$$\& R'_{C_1} > R'_{C_2}$$

For current source $P \propto R$

$$Q_1 > Q_2 \text{ (Option (D) also correct)}$$

7. Ans (C,D)



$$P_{\text{gas}} = P_a + \frac{4S}{r}$$

$$PV^\gamma = \text{constant [adiabatic process]}$$

$$\left(P_{a_1} + \frac{4S}{r_1}\right) \left(\frac{4}{3}\pi r_1^3\right)^{5/3} = \left(P_{a_2} + \frac{4S}{r_2}\right) \left(\frac{4}{3}\pi r_2^3\right)^{5/3}$$

$$\frac{r_1^3}{r_2^3} = \left(\frac{P_{a_2} + \frac{4S}{r_2}}{P_{a_1} + \frac{4S}{r_1}}\right)$$

$$P^{1-\gamma} T^\gamma = \text{constant}$$

$$\left(P_{a_2} + \frac{4S}{r_2}\right)^{1-5/3} T_2^{5/3} = \left(P_{a_1} + \frac{4S}{r_1}\right)^{1-5/3} T_1^{5/3}$$

$$\left(\frac{T_2}{T_1}\right)^{5/3} = \left(\frac{P_{a_1} + \frac{4S}{r_1}}{P_{a_2} + \frac{4S}{r_2}}\right)^{-2/3}$$

$$\left(\frac{T_2}{T_1}\right)^{5/2} = \left(\frac{P_{a_2} + \frac{4S}{r_2}}{P_{a_1} + \frac{4S}{r_1}}\right)$$

(D) is correct

8. Ans (A,C,D)

$$W_{el} + W_{ext} = k_f - k_i$$

$$qV_i - qV_f + W_{ext} = k_f - k_i$$

$$\frac{q\sigma}{2\epsilon_0} \left[\sqrt{R^2 + Z^2} - Z \right] - \frac{q\sigma R}{2\epsilon_0} + CZ = k_f - 0$$

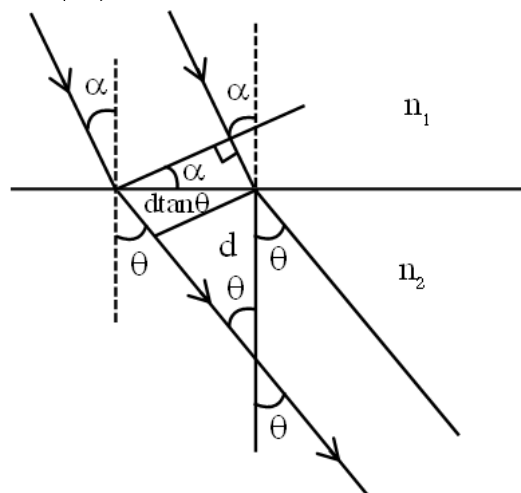
$$C = \frac{q\sigma B}{2\epsilon_0}$$

Substitute β & Z , calculate kinetic energy at $z = 0$.

If kinetic energy is positive, then particle will reach at origin.

If kinetic energy is negative, then particle will not reach at origin.

9. Ans (B)



Optical path difference $\rightarrow$

$$\Delta x = n_1(d \tan \theta) \sin \alpha - n_2(d \tan \theta) \sin \theta$$

$$= (n_1 \sin \alpha - n_2 \sin \theta) d \tan \theta$$

$$= 0$$

$$\Rightarrow \Delta \phi = 0$$

10. Ans (A,B,C,D)

For adiabatic process (A $\rightarrow$ B)

$$P_A V_A^\gamma = P_B V_B^\gamma$$

$$10^5 \times (0.8)^{\frac{5}{3}} = 3 \times 10^5 (V_B)^{\frac{5}{3}}$$

$$\Rightarrow V_B = 0.8 \times \left(\frac{1}{3}\right)^{0.6} = 0.4$$

Work done in process A $\rightarrow$ B

$$W_{AB} = \frac{P_A V_A - P_B V_B}{\gamma - 1}$$

$$\Rightarrow W_{AB} = \frac{10^5 \times 0.8 - 3 \times 10^5 \times 0.4}{\frac{5}{3} - 1}$$

$$\Rightarrow W_{AB} = -60 \text{ kJ} \Rightarrow |W_{AB}| = 60 \text{ kJ}$$

Work done in process B $\rightarrow$ C (Isothermal process)

$$W_{BC} = nRT \ln \frac{V_C}{V_B} = P_B V_B \ln \frac{V_C}{V_B}$$

$$\Rightarrow W_{BC} = 3 \times 10^5 \times 0.4 \ln \frac{0.8}{0.4}$$

$$\Rightarrow W_{BC} = 84 \text{ kJ}$$

Work done in process C $\rightarrow$ A

$$W_{CA} = P \Delta V = 0 \quad (\because \Delta V = 0)$$

So total work done in the process A $\rightarrow$ B $\rightarrow$ C

$$W_{ABC} = W_{AB} + W_{BC} + W_{CA} = -60 + 84 + 0$$

$$W_{ABC} = 24 \text{ kJ}$$

PART-1 : PHYSICS

SECTION-III

1. Ans (1)

Torque about origin is zero. So angular momentum about origin remains conserved.

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{1}{\sqrt{2}} & -\sqrt{2} & 0 \\ -\sqrt{2} & \sqrt{2} & \frac{2}{\pi} \end{vmatrix} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & 0.5 \\ v_x & v_y & \frac{2}{\pi} \end{vmatrix}$$

$$\hat{i} \left[\sqrt{2} \times \frac{2}{\pi} \right] - \hat{j} \left[\frac{\sqrt{2}}{\pi} \right] + \hat{k} [1 - 2]$$

$$= \hat{i} \left[\frac{y \times 2}{\pi} - 0.5 v_y \right] - \hat{j} \left[\frac{x \times 2}{\pi} - 0.5 v_x \right] + \hat{k} [x v_y - y v_x]$$

$$x v_y - y v_x = x - 1$$

2. Ans (8)

$$\text{Th}_{90}^{230} \rightarrow \text{Po}_{84}^{214} + n \alpha + m \beta^-$$

$$230 = 214 + 4n$$

$$n = \frac{16}{4} = 4$$

$$90 = 84 + n \times 2 - m \times 1$$

$$90 = 84 + 4 \times 2 - m \times 1$$

$$m = 92 - 90 = 2$$

$$\text{Hence, } n \times m = 4 \times 2 = 8$$

3. Ans (6)

For the balanced Wheatstone bridge

$$\frac{R_1}{R_2} = \frac{\int_0^{0.5} \frac{\rho dx}{\pi r_x^2}}{\int_{0.5}^1 \frac{\rho dx}{\pi r_x^2}}$$

$$\frac{R_1}{R_2} = \frac{\left[\frac{1}{r_x} \right]_0^{0.5}}{\left[\frac{1}{r_x} \right]_{0.5}^1}$$

$$\therefore R_1 = 5R_2 = 6\Omega$$

4. **Ans (0)**

$$B = e^{\alpha} (m_e)^{\beta} h^{\gamma} k^{\delta}$$

$$[B] = [e^{\alpha}] [m_e]^{\beta} [h]^{\gamma} [k]^{\delta}$$

$$[M^1 T^{-2} A^{-1}] = [AT]^{\alpha} [m]^{\beta} [ML^2 T^{-1}]^{\gamma} [ML^3 A^{-2} T^{-4}]^{\delta}$$

$$M^1 T^{-2} A^{-1} = m^{\beta + \gamma + \delta} L^{2\gamma + 3\delta} T^{\alpha - \gamma - 4\delta} A^{\alpha - 2\delta}$$

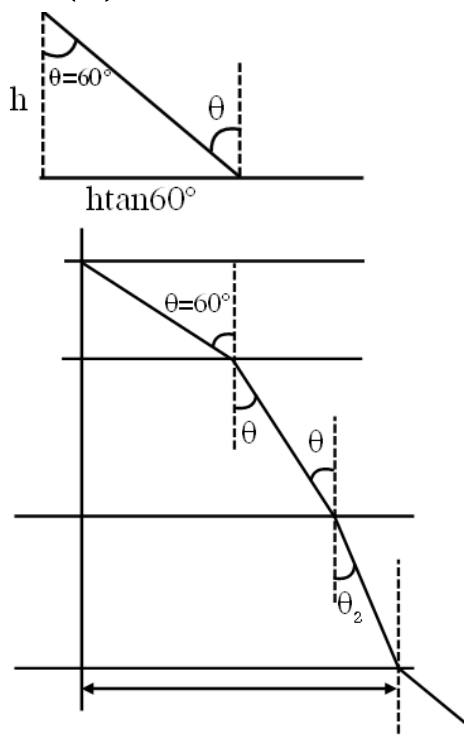
$$\text{Compare : } \beta + \gamma + \delta = 1; 2\gamma + 3\delta = 0,$$

$$\alpha - \gamma - 4\delta = -2, \alpha - 2\delta = -1$$

$$\text{On solving } \alpha = 3, \beta = 2, \gamma = -3, \delta = 2$$

$$\alpha - \beta + \gamma + \delta = 0$$

5. **Ans (6)**



$$1 \sin 60^\circ = \sqrt{\frac{3}{2}} \sin \theta \Rightarrow \theta_1 = 45^\circ$$

$$\sqrt{\frac{3}{2}} \sin 45^\circ = \sqrt{3} \sin \theta_2$$

$$= \sqrt{\frac{3}{2}} \frac{1}{\sqrt{2}} = \sqrt{3} \sin \theta_2 = \theta_2 = 30^\circ$$

$$h \tan 60^\circ + d \tan 45^\circ + d \tan 30^\circ$$

$$\frac{1}{3} \sqrt{3} + \left(\frac{\sqrt{3} - 1}{2} \right) + \left(\frac{\sqrt{3} - 1}{2} \right) \frac{1}{\sqrt{3}}$$

$$\frac{2\sqrt{3} + 3\sqrt{3} - 3 + 3 - \sqrt{3}}{6}$$

$$\frac{4\sqrt{3}}{6}$$

$$\therefore n \frac{4\sqrt{3}}{6} = 4\sqrt{3}$$

$$n = 6$$

6. **Ans (2)**

From Gauss law,

$$\phi_{\text{hemisphere}} + \phi_{\text{Cone}} = \frac{q}{\epsilon_0} \quad \dots\dots\dots(i)$$

Total flux produced from q in α angle

$$\phi = \frac{q}{2\epsilon_0} [1 - \cos \alpha]$$

$$\text{For hemisphere, } \alpha = \frac{\pi}{2}$$

$$\phi_{\text{hemisphere}} = \frac{q}{2\epsilon_0}$$

From equation (i)

$$= \frac{q}{2\epsilon_0} + \phi_{\text{cone}} = \frac{q}{\epsilon_0}$$

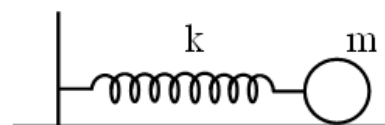
$$\phi_{\text{cone}} = \frac{q}{2\epsilon_0}$$

$$\frac{4q}{6\epsilon_0} = \frac{q}{2\epsilon_0}$$

$$n = 3$$

Alternatively, $\phi \propto$ no of electric field lines passing through surface q is point charge which has uniformly distributed electric field lines thus half of electric field lines will pass through hemisphere & other half will pass through conical surface.

7. **Ans (8)**



$$\ell > \ell_0 \rightarrow k = k_1$$

$$\ell < \ell_0 \rightarrow k = k_2$$

Time period of oscillation,

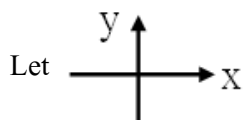
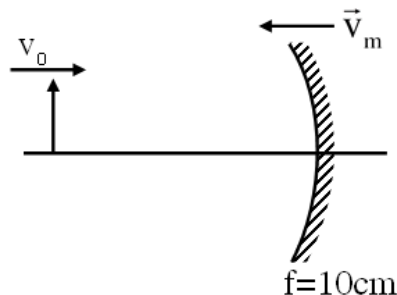
$$T = \pi \sqrt{\frac{m}{k_1}} + \pi \sqrt{\frac{m}{k_2}}$$

$$T = \pi \sqrt{\frac{0.1}{0.009}} + \pi \sqrt{\frac{0.1}{0.004}}$$

$$T = \frac{\pi}{0.3} + \frac{\pi}{0.2} \Rightarrow T = 8.33\pi$$

$$\text{So, } n = 8$$

8. Ans (4)



$$u = -30 \text{ cm}$$

$$f = -10 \text{ cm}$$

$$v = \frac{f_0}{u - f} = -15 \text{ cm}$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{du}{dt} = -\frac{v^2}{u^2} \frac{du}{dt}$$

$$\vec{v}_{Im} = -\left(\frac{v}{u}\right)^2 \vec{v}_{om}$$

$$\text{Given } \vec{v}_l = \vec{0}$$

$$\vec{v}_l - \vec{v}_m = -\left(\frac{-15}{-30}\right)^2 (\vec{v}_{O/m})$$

$$\vec{v}_l - \vec{v}_m = -\frac{1}{4} \vec{v}_0 + \frac{1}{4} \vec{v}_m$$

$$\vec{v}_0 = 20 \text{ cm/s } \hat{i}$$

$$\vec{v}_l = \vec{0} \text{ cm/s}$$

$$\frac{5}{4} \vec{v}_m = \frac{\vec{v}_0}{4}$$

$$\vec{v}_m = \frac{\vec{v}_0}{4} = \frac{20 \text{ cm/s } \hat{i}}{5} = 4 \text{ m/s } \hat{i}$$

$$|\vec{v}_m|_{\text{cm/s}} = 4$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (C)

Atom 'X' occupies FCC lattice points as well as alternate tetrahedral voids of the same lattice

$$\Rightarrow \frac{1}{4} \text{th distance of body diagonal}$$

$$= \frac{\sqrt{3}a}{4} = 2r_X$$

$$\Rightarrow a = \frac{8r_X}{\sqrt{3}}$$

Number of atoms of X per unit cell

$$= 4 + 8 = 12$$

(FCC lattice points) (tetrahedral voids)

% packing efficiency =

$$\frac{\text{Volume occupied by X}}{\text{Volume of cubic unit cell}} \times 100$$

$$= \frac{12 \times \frac{4}{3} \pi (r_X)^3}{a^3} \times 100$$

$$= \frac{12 \times \frac{4}{3} \pi (r_X)^3}{\left(\frac{8r_X}{\sqrt{3}}\right)^3} \times 100$$

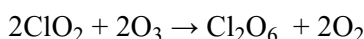
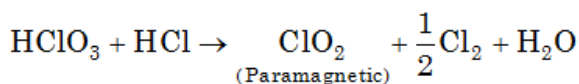
$$= \left(12 \times \frac{4}{3} \times \pi \times \frac{1}{8^3} \times 3\sqrt{3}\right) \times 100$$

$$= \frac{3\sqrt{3}\pi}{32} \times 100$$

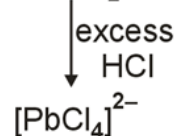
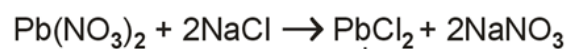
$$= 50.9\%$$

Hence, option (C) is the most appropriate option

2. Ans (C)

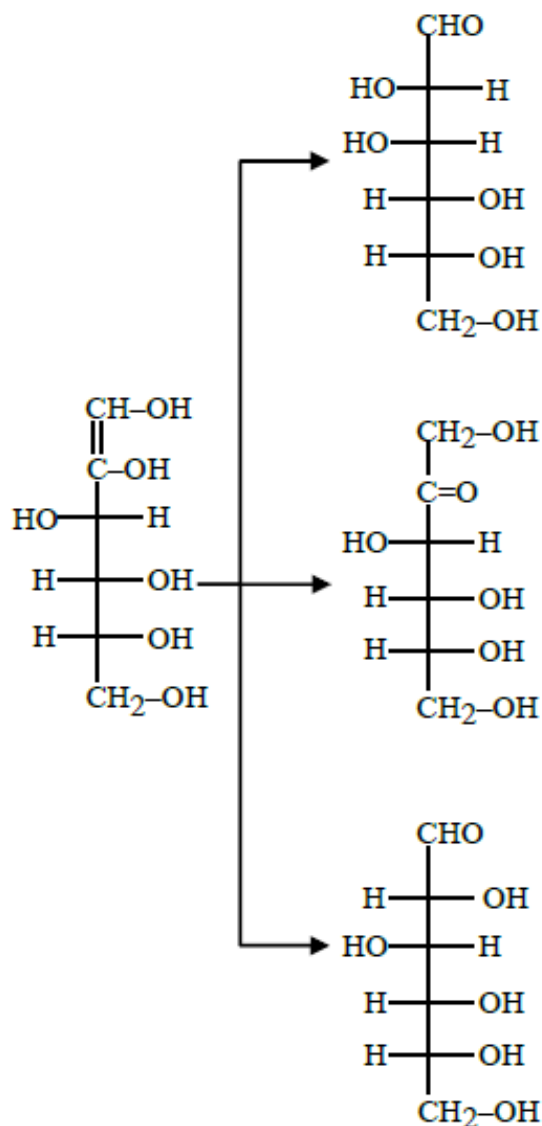


3. Ans (C)



4. Ans (A)

Basic catalyse tautomerism through enediol intermediate



PART-2 : CHEMISTRY

SECTION-I (ii)

5. Ans (A,C,D)

Compound	Weight % of P	Weight % of Q
1	50	50
2	44.4	55.6
3	40	60

For option (A)

Let atomic mass of P be M_P and atomic mass of Q be M_Q

Molar ratio of atoms P : Q in compound 3 is

$$\frac{40}{M_P} : \frac{60}{M_Q} = 3 : 4$$

$$\frac{2M_Q}{3M_P} = \frac{3}{4} \Rightarrow 9M_P = 8M_Q$$

Molar ratio of atoms P : Q in compound 2 is

$$\frac{44.4}{M_P} : \frac{55.6}{M_Q}$$

$$= 44.4 M_Q : 55.6 M_P$$

$$= 44.4 M_Q : 55.6 \times \frac{8M_Q}{9}$$

$$= 44.4 : 55.6 \times \frac{8}{9}$$

$$= 9 : 10$$

$\Rightarrow$ Empirical formula of compound 2 is therefore

$$P_9Q_{10}$$

Option (A) is correct

For option (B)

Molar Ratio of atoms P : Q in compound 3 is

$$\frac{40}{M_P} : \frac{60}{M_Q} = 3 : 2$$

$$\frac{2M_Q}{3M_P} = \frac{3}{2} \Rightarrow 9M_P = 4M_Q$$

$$\text{If } M_P = 20 \Rightarrow M_Q = \frac{9 \times 20}{4} = 45$$

Option (B) is incorrect

For option (C)

Molar ratio of atoms P : Q in compound 2 is

$$\frac{44.4}{M_P} : \frac{55.6}{M_Q} = 44.4M_Q : 55.6 M_P = 1 : 1$$

$$\Rightarrow \frac{M_P}{M_Q} = \frac{44.4}{55.6}$$

Molar ratio of atoms P : Q in compound 1 is

$$\frac{50}{M_P} : \frac{50}{M_Q} = M_Q : M_P$$

$$= 55.6 : 44.4$$

$$\simeq 5 : 4$$

Hence, empirical formula of compound 1 is P_5Q_4

Hence, option (C) is correct

For option (D)

Molar ratio of atoms P : Q in compound 1 is

$$\frac{50}{M_P} : \frac{50}{M_Q} = M_Q : M_P$$

$$= 35 : 70 = 1 : 2$$

Hence, empirical formula of compound 1 is PQ_2

Hence, option (D) is correct

6. **Ans (C,D)**

$$\Delta G = \Delta H - T\Delta S$$

$$\Delta G = \Delta H + T \left(\frac{d\Delta G}{dT} \right)_p$$

$$-nF \left(\frac{dE_{\text{cell}}}{dT} \right) = -\Delta S$$

$$\frac{dE_{\text{cell}}}{dT} = \frac{\Delta S}{nF} = \frac{2R}{3F} \text{ (given)}$$

$$\Rightarrow \Delta S = 2nR/3$$

For the reaction, $M(g) + 3H^+(aq) \longrightarrow (3/2)H_2(g) + M^{3+}(aq)$

$$n = 3$$

$$\Rightarrow \Delta S = 2R$$

Hence, option (A) is incorrect

For the reaction, $Pt_{(s)} | H_{2(g)}, 1 \text{ bar} | H^+_{aq}(0.1M) || H^+_{aq}(0.01M) | H_{2(g)}, 1 \text{ bar} | Pt_{(s)}$

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0591}{1} \log \frac{0.1}{0.01} = -0.0591V$$

E_{cell} is negative $\Rightarrow \Delta G > 0$ and $\Delta S < 0$ ($\Delta H = 0$ for concentration cells)

Hence, option (B) is incorrect

Racemization of an optically active compound is a spontaneous process.

Here, $\Delta H = 0$ (similar type of bonds are present in enantiomers)

$$\Rightarrow \Delta S > 0$$

Hence, option (C) is correct.

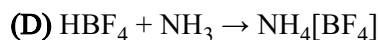
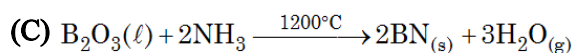
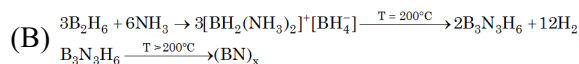
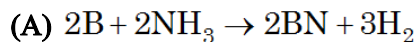
$[CoCl_6]^{4-} + 3 \text{ en} \rightarrow [Co(en)_3]^{2+} + 6Cl^-$ is a spontaneous process

more stable complex is formed

$$\Rightarrow \Delta S > 0$$

Hence, option (D) is correct.

7. **Ans (A,C)**



8. **Ans (A,C,D)**

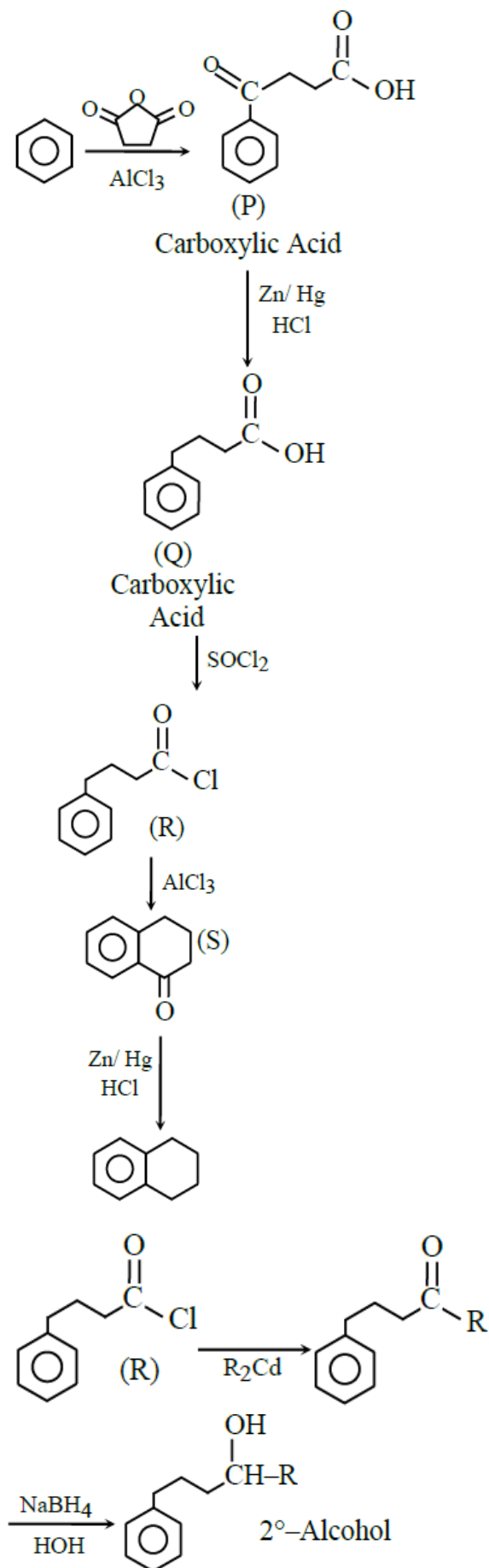
(A) $CaO + SiO_2 \rightarrow CaSiO_3$ (in the temperature range 900 – 1500 K)

(B) In fusion zone molten iron becomes heavy by absorbing elemental impurities and produces Pig iron. (in the temperature range 900 – 1500 K) which contains 4% carbon

(C) $C + CO_2 \rightarrow 2CO$ (in the temperature range 900 – 1500 K)

(D) Exhaust gases contains CO and CO_2

9. Ans (A,C,D)



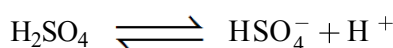
10. Ans (A,B,C)

- (a) The polymerisation of neoprene gives natural rubber.
 (b) is correct statement
 (c) is correct statement
 (d) Ethene at 350-570 K temperature and 1000-2000 atm pressure in the presence of a peroxide initiator yields low density polythene.

PART-2 : CHEMISTRY

SECTION-III

1. Ans (2)



1M

-

Na_2SO_4

$2 \times 10^{-2} \text{ M}$

$2 \times 10^{-2} \text{ M}$

$2 \times 10^{-2} \text{ M}$

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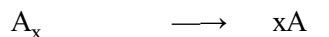
2. Ans (5)

0.1 mole ionic salt in 0.9 kg water at 35° C

Vapour pressure of solution = 59.500 mm of Hg

Vapour pressure of pure H₂O = 60.000 mm of Hg

Let the number of ions present per formula unit of the ionic salt be 'x'



(Salt) (Ions)

0.1 -

0.1 (1 - 0.8) (0.1 × 0.8) x

Total moles of non-volatile particles = 0.02 + 0.08 x
in 0.9 kg water

$$\text{Moles of water} = \frac{0.9 \times 10^3}{18} = 50 \text{ moles}$$

$$\text{Relative lowering of vapour pressure} = \frac{P^\circ - P_s}{P^\circ}$$

= Mole fraction of non - volatile particles

$$\frac{P^\circ - P_s}{P_s} = \frac{\text{moles of non - volatile particles}}{\text{moles of water}}$$

$$\frac{60.000 - 59.500}{59.500} = \frac{0.02 + 0.08x}{50}$$

$$x = 5$$

3. Ans (8)

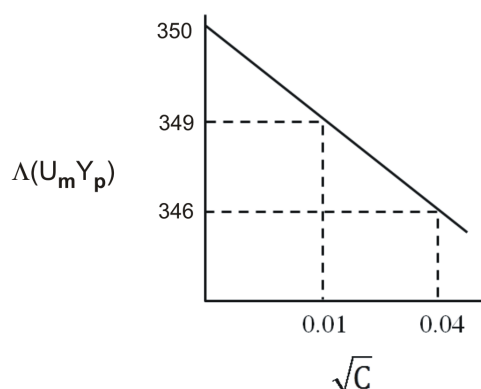
$$\Lambda^\circ(Z_m X_n) = m \times \lambda_{Z^{n+}}^\circ + n \times \lambda_{X^{m-}}^\circ = 250$$

$$5m + 8n = 25 \quad \dots\dots(1)$$

$$\Lambda^\circ(V_m X_n) = m \times \lambda_{V^{n+}}^\circ + n \times \lambda_{X^{m-}}^\circ = 440$$

$$100m + 80n = 440$$

$$5m + 4n = 22 \quad \dots\dots(2)$$



From the extrapolation of curve

$$\Lambda^\circ(U_m Y_p) = 350$$

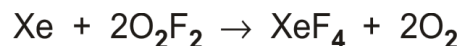
$$m \times \lambda_{U^{p+}}^\circ + p \times \lambda_{Y^{m-}}^\circ = 350$$

$$25m + 100p = 350$$

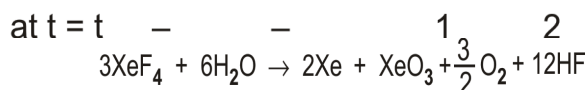
$$m + 4p = 14 \quad \dots\dots(3)$$

Solving we get (m + n + p) = 8

4. Ans (9)



at t = 0 1 2 - -



at t = 0 1 2 - - -

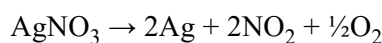
at t = t - - $\frac{2}{3}$ $\frac{1}{3}$ $\frac{1}{2}$ 4

$$X = 4 \text{ mol}$$

$$Y = 2.5 \text{ mol}$$

$$X + 2Y = 9$$

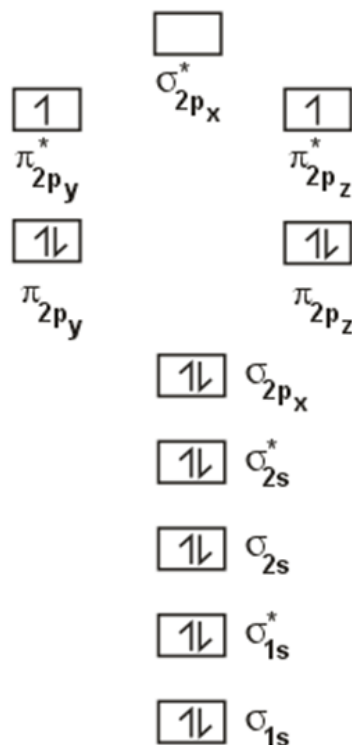
5. Ans (5)



- Both NO₂ & O₂ are paramagnetic

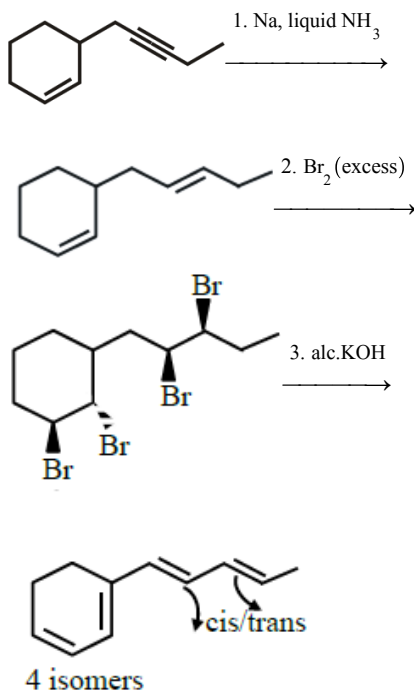
- NO₂ is odd electron molecule with one unpaired electron (five lone pairs)

-O₂ has two unpaired electrons

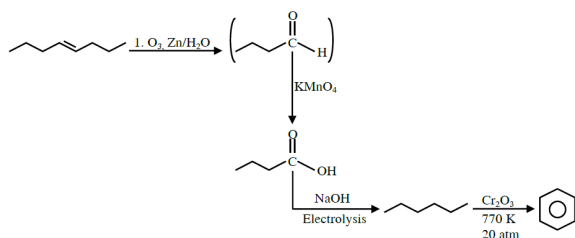


Total number of bonding electrons = 10

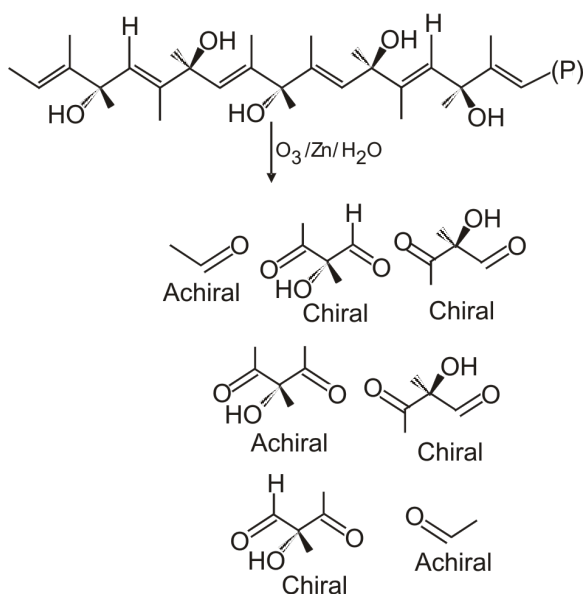
6. Ans (4)



7. Ans (6)



8. Ans (4)



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (D)

3R	3R	3R	3R
2B	2B	2B	2B

B-1 B-2 B-3 B-4

Case-I : when exactly one box provides four balls (3R 1B or 2R 2B)

Number of ways in this case ${}^5\text{C}_4 ({}^3\text{C}_1 \times {}^2\text{C}_1)^3 \times 4$

Case-II : when exactly two boxes provide three balls (2R 1B or 1R 2B) each

Number of ways in this case $({}^5\text{C}_3 - 1)^2$
 $({}^3\text{C}_1 \times {}^2\text{C}_1)^2 \times 6$

Required number of ways = 21816

2. Ans (B)

$$M = \begin{bmatrix} \frac{5}{2} & \frac{3}{2} \\ -\frac{3}{2} & -\frac{1}{2} \end{bmatrix}$$

$$M = \begin{bmatrix} \frac{3}{2} + 1 & \frac{3}{2} \\ -\frac{3}{2} & -\frac{3}{2} + 1 \end{bmatrix}$$

$$M = I + \frac{3}{2} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$$

$$\text{Let } A = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$M^{2022} = \left(I + \frac{3}{2} A \right)^{2022}$$

$$= I + 3033A$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + 3033 \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 3034 & 3033 \\ -3033 & -3032 \end{bmatrix}$$

3. Ans (D)

Box I 8(R) 3(B) 5(G)

Box II 24(R) 9(B) 15(G)

Box III 1(B) 12(G) 3(y)

Box IV 10(G) 16(o) 6(w)

A (one of the chosen balls is white)

B (at least one of the chosen ball is green)

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$$

$$A \cap B \rightarrow (wG)$$

$$\begin{aligned} &= \frac{\frac{5}{16} \times \frac{6}{32}}{\frac{5}{16} \times 1 + \frac{8}{16} \times \frac{15}{48} + \frac{3}{16} \times \frac{12}{16}} \\ &= \frac{15}{156} = \frac{5}{52} \end{aligned}$$

4. Ans (B)

$$f(n) = n + \sum_{r=1}^n \frac{16r + (9 - 4r)n - 3n^2}{4rn + 3n^2}$$

$$f(n) = n + \sum_{r=1}^n \frac{(16r + 9n) - (4rn + 3n^2)}{4rn + 3n^2}$$

$$f(n) = n + \left(\sum_{r=1}^n \frac{16r + 9n}{4rn + 3n^2} \right) - n$$

$$\lim_{n \rightarrow \infty} f(n) = \lim_{n \rightarrow \infty} \sum \frac{16r + 9n}{4rn + 3n^2}$$

$$= \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{(16(\frac{r}{n}) + 9) \frac{1}{n}}{4(\frac{r}{n}) + 3}$$

$$= \int_0^1 \frac{16x + 9}{4x + 3} dx = \int_0^1 4 dx - \int_0^1 \frac{3 dx}{4x + 3}$$

$$= 4 - \frac{3}{4} (\ln |4x + 3|)_0^1$$

$$= 4 - \frac{3}{4} \ln \frac{7}{3}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (D)

$$\text{Let } (\bar{z})^2 + \frac{1}{z^2} = m + in, m, n \in \mathbb{Z}$$

$$(\bar{z})^2 + \frac{\bar{z}^2}{|z|^4} = m + in$$

$$\Rightarrow (x^2 - y^2) \left(1 + \frac{1}{|z|^4} \right) = m \quad \dots(1)$$

$$\& -2xy \left(1 + \frac{1}{|z|^4} \right) = n \quad \dots(2)$$

$$\text{Equation (1)}^2 + (2)^2$$

$$\left(1 + \frac{1}{|z|^4} \right)^2 \left[(x^2 + y^2)^2 \right] = m^2 + n^2$$

$$\left(1 + \frac{1}{|z|^4} \right)^2 (|z|^4)^2 = m^2 + n^2$$

$$\Rightarrow |z|^4 + \frac{1}{|z|^4} + 2 = m^2 + n^2$$

Now for option (A)

$$|z|^4 = \frac{43 + 3\sqrt{205}}{2}$$

$$\Rightarrow m^2 + n^2 = 45$$

$$\Rightarrow m = \pm 6, n = \pm 3$$

Option (B)

$$|z|^4 + \frac{1}{|z|^4} + 2 = \frac{7 + \sqrt{33}}{4} + \frac{7 - \sqrt{33}}{4} + 2 = \frac{7}{2} + 2 = \frac{11}{2}$$

Option (C)

$$|z|^4 + \frac{1}{|z|^4} + 2 = \frac{9 + \sqrt{65}}{4} + \frac{9 - \sqrt{65}}{4} + 2 = \frac{18}{4} + 2 = \frac{9}{2} + 2 = \frac{13}{2}$$

Option (D)

$$|z|^4 + \frac{1}{|z|^4} + 2 = \frac{7 + \sqrt{13}}{6} + \frac{7 - \sqrt{13}}{6} + 2 = \frac{14}{6} + 2 = \frac{7}{3} + 2 = \frac{13}{2}$$

6. Ans (B,D)

$$\vec{a} = 3\hat{i} + \hat{j} - \hat{k}$$

$$\vec{b} = \hat{i} + b_2\hat{j} + b_3\hat{k}$$

$$\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$$

$$\begin{pmatrix} 0 & -c_3 & c_2 \\ c_3 & 0 & -c_1 \\ -c_2 & c_1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} 3 - c_1 \\ 1 - c_2 \\ -1 - c_3 \end{pmatrix}$$

multiply & compare

$$b_2c_3 - b_3c_2 = c_1 - 3 \quad \dots(1)$$

$$c_3 - b_3c_1 = 1 - c_2 \quad \dots(2)$$

$$c_2 - b_2c_1 = 1 + c_3 \quad \dots(3)$$

$$(1)\hat{i} - (2)\hat{j} + (3)\hat{k}$$

$$\hat{i}(b_2c_3 - b_3c_2) - \hat{j}(c_3 - b_3c_1) + \hat{k}(c_2 - b_2c_1)$$

$$= c_1\hat{i} + c_2\hat{j} + c_3\hat{k} - 3\hat{i} - \hat{j} + \hat{k}$$

$$\vec{b} \times \vec{c} = \vec{c} - \vec{a}$$

Take dot product with $\vec{b}$

$$0 = \vec{c} \cdot \vec{b} - \vec{a} \cdot \vec{b}$$

$$\vec{b} \cdot \vec{c} = 0$$

$$\vec{b} \perp \vec{c}$$

$$\vec{b} \wedge \vec{c} = 90^\circ$$

Take dot product with $\vec{c}$

$$0 = |\vec{c}|^2 - \vec{a} \cdot \vec{c}$$

$$\vec{a} \cdot \vec{c} = |\vec{c}|^2$$

$$\vec{a} \cdot \vec{c} \neq 0$$

$$\vec{b} \times \vec{c} = \vec{c} - \vec{a}$$

Squaring

$$|\vec{b}|^2 |\vec{c}|^2 = |\vec{c}|^2 + |\vec{a}|^2 - 2\vec{c} \cdot \vec{a}$$

$$|\vec{b}|^2 |\vec{c}|^2 = |\vec{c}|^2 + 11 - 2|\vec{c}|^2$$

$$|\vec{b}|^2 |\vec{c}|^2 = 11 - |\vec{c}|^2$$

$$|\vec{c}|^2 (|\vec{b}|^2 + 1) = 11$$

$$|\vec{c}|^2 = \frac{11}{|\vec{b}|^2 + 1}$$

$$|\vec{c}| \leq \sqrt{11}$$

$$\text{given } \vec{a} \cdot \vec{b} = 0$$

$$b_2 - b_3 = -3 \quad \text{also}$$

$$b_2^2 + b_3^2 - 2b_2b_3 = 9 \quad b_2b_3 > 0$$

$$b_2^2 + b_3^2 = 9 + 2b_2b_3$$

$$b_2^2 + b_3^2 = 9 + 2b_2b_3 > 9$$

$$b_2^2 + b_3^2 > 9$$

$$|\vec{b}| = \sqrt{1 + b_2^2 + b_3^2}$$

$$|\vec{b}| > \sqrt{10}$$

7. Ans (A)

$$\frac{dy}{dx} + 12y = \cos\left(\frac{\pi}{12}x\right)$$

Linear D.E.

$$\text{I.F.} = e^{\int 12 \cdot dx} = e^{12x}$$

Solution of DE

$$y \cdot e^{12x} = \int e^{12x} \cdot \cos\left(\frac{\pi}{12}x\right) dx$$

$$y \cdot e^{12x} = \frac{e^{12x}}{(12)^2 + \left(\frac{\pi}{12}\right)^2} \left(12 \cos \frac{\pi}{12}x + \frac{\pi}{12} \sin \frac{\pi}{12}x\right) + C$$

$$\Rightarrow y = \frac{(12)}{(12)^4 + \pi^2} \left((12)^2 \cos\left(\frac{\pi x}{12}\right) + \pi \sin\left(\frac{\pi x}{12}\right)\right) + \frac{C}{e^{12x}}$$

Given $y(0) = 0$

$$\Rightarrow 0 = \frac{12}{12^4 + \pi^2} (12^2 + 0) + C \Rightarrow C = \frac{-12^3}{12^4 + \pi^2}$$

$$\therefore y = \frac{12}{12^4 + \pi^2} \left[(12)^2 \cos\left(\frac{\pi x}{12}\right) + \pi \sin\left(\frac{\pi x}{12}\right) - 12^3 \cdot e^{-12x}\right]$$

$$\text{Now } \frac{dy}{dx} = \frac{12}{12^4 + \pi^2} \left[\underbrace{-12\pi \sin\left(\frac{\pi x}{12}\right) + \frac{\pi^2}{12} \cos\left(\frac{\pi x}{12}\right)}_{\text{min. value}} + 12^3 e^{-12x} \right]$$

$$\left(-\sqrt{144\pi^2 + \frac{\pi^4}{144}} = -12\pi \sqrt{1 + \frac{\pi^2}{12^4}} \right)$$

$$\Rightarrow \frac{dy}{dx} > 0 \quad \forall x \leq 0 \text{ \& may be negative/positive}$$

for $x > 0$

So, $f(x)$ is neither increasing nor decreasing

For some $\beta \in \mathbb{R}$, $y = \beta$ intersects $y = f(x)$ at infinitely many points

8. **Ans (C)**

$$\int_1^e \frac{(\log_e x)^{1/2}}{x(a - (\log_e x)^{3/2})^2} dx = 1$$

$$\text{Let } a - (\log_e x)^{3/2} = t$$

$$\frac{(\log_e x)^{1/2}}{x} dx = -\frac{2}{3} dt$$

$$= \frac{2}{3} \int_a^{a-1} \frac{-dt}{t^2} = \frac{2}{3} \left(\frac{1}{t} \right)_a^{a-1} = 1$$

$$\frac{2}{3a(a-1)} = 1$$

$$3a^2 - 3a - 2 = 0$$

$$a = \frac{3 \pm \sqrt{33}}{6}$$

9. **Ans (A,B,C)**

$$a_1 = 7, d = 8$$

$$T_{n+1} - T_n = a_n \forall n \geq 1$$

$$S_n = T_1 + T_2 + T_3 + \dots + T_{n-1} + T_n$$

$$S_n = T_1 + T_2 + T_3 + \dots + T_{n-1} + T_n$$

on subtraction

$$T_n = T_1 + a_1 + a_2 + \dots + a_{n-1}$$

$$T_n = 3 + (n-1)(4n-1)$$

$$T_n = 4n^2 - 5n + 4$$

$$\sum_{k=1}^n T_k = 4 \sum_{k=1}^n k^2 - 5 \sum_{k=1}^n k + 4n$$

$$T_{20} = 1504$$

$$T_{30} = 3454$$

$$\sum_{k=1}^{30} T_k = 35615$$

$$\sum_{k=1}^{20} T_k = 10510$$

10. **Ans (A,B)**

$$\text{line of intersection is } \frac{x}{0} = \frac{y-4}{-4} = \frac{z}{5}$$

(1) Any skew line with the line of intersection of given planes can be edge of tetrahedron.

(2) any intersecting line with line of intersection of given planes must lie either in plane P_1 or P_2 can be edge of tetrahedron.

PART-3 : MATHEMATICS

SECTION-III

1. **Ans (4)**

$$\alpha \in \left(0, \frac{\pi}{4}\right), \beta \in \left(-\frac{\pi}{4}, 0\right) \Rightarrow \alpha + \beta \in \left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$$

$$\sin(\alpha + \beta) = \frac{1}{3}, \cos(\alpha - \beta) = \frac{2}{3}$$

$$\left(\frac{\sin \alpha}{\cos \beta} + \frac{\cos \alpha}{\sin \beta} + \frac{\cos \beta}{\sin \alpha} + \frac{\sin \beta}{\cos \alpha} \right)^2$$

$$\left(\frac{\cos(\alpha - \beta)}{\cos \beta \sin \beta} + \frac{\cos(\beta - \alpha)}{\sin \alpha \cos \alpha} \right)^2$$

$$= 4\cos^2(\alpha - \beta) \left(\frac{1}{\sin 2\beta} + \frac{1}{\sin 2\alpha} \right)^2$$

$$= 4\cos^2(\alpha - \beta) \left(\frac{2\sin(\alpha + \beta)\cos(\alpha - \beta)}{\sin 2\alpha \sin 2\beta} \right) \dots (1)$$

$$= \frac{16\cos^4(\alpha - \beta)\sin^2(\alpha + \beta) \times 4}{(\cos 2(\alpha - \beta) - \cos 2(\alpha + \beta))^2}$$

$$= \frac{64\cos^4(\alpha - \beta)\sin^2(\alpha + \beta)}{(2\cos^2(\alpha - \beta) - 1 - 1 + 2\sin^2(\alpha + \beta))^2}$$

$$= 64 \times \frac{16}{81} \times \frac{1}{9} \frac{1}{\left(2 \times \frac{4}{9} - 1 - 1 + \frac{2}{9}\right)^2}$$

$$= \frac{64 \times 16}{81 \times 9} \cdot \frac{81}{64} = \frac{16}{9}$$

2. **Ans (9)**

$$f(x) = \log_2(x^3 + 1) = y$$

$$x^3 + 1 = 2^y \Rightarrow x = (2^y - 1)^{1/3} = f^{-1}(y)$$

$$f^{-1}(x) = (2^x - 1)^{1/3}$$

$$= \int_1^2 \log_2(x^3 + 1) dx + \int_1^{\log_2 9} (2^x - 1)^{1/3} dx$$

$$= \int_1^2 f(x) dx + \int_1^{\log_2 9} f^{-1}(x) dx = 2\log_2 9 - 1$$

3. Ans (1)

$$x^{16(\log_5 x)^3 - 68 \log_5 x} = 5^{-16}$$

Take log to the base 5 on both sides and put

$$\log_5 x = t$$

$$16t^4 - 68t^2 + 16 = 0$$

$$\Rightarrow 4t^4 - 17t^2 + 4 = 0 \quad \begin{cases} t_1 \\ t_2 \\ t_3 \\ t_4 \end{cases}$$

$$t_1 + t_2 + t_3 + t_4 = 0$$

$$\log_5 x_1 + \log_5 x_2 + \log_5 x_3 + \log_5 x_4 = 0$$

$$x_1 x_2 x_3 x_4 = 1$$

4. Ans (4)

$$A = \begin{pmatrix} \beta & 0 & 1 \\ 2 & 1 & -2 \\ 3 & 1 & -2 \end{pmatrix} \quad |A| = -1$$

$$\Rightarrow |A^7 - (\beta - 1)A^6 - \beta A^5| = 0$$

$$\Rightarrow |A|^5 |A^2 - (\beta - 1)A - \beta I| = 0$$

$$\Rightarrow |A|^5 |(A^2 - \beta A) + A - \beta I| = 0$$

$$\Rightarrow |A|^5 |A(A - \beta I) + I(A - \beta I)| = 0$$

$$|A|^5 |(A + I) (A - \beta I)| = 0$$

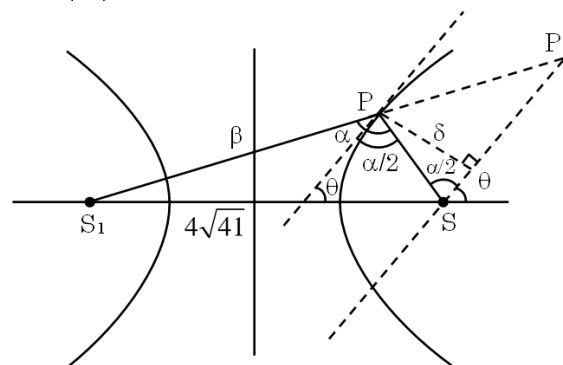
$$A + I = \begin{pmatrix} \beta + 1 & 0 & 1 \\ 2 & 2 & -2 \\ 3 & 1 & -1 \end{pmatrix} \Rightarrow |A + I| = -4$$

, Here $|A| \neq 0$ & $|A + I| \neq 0$

$$A - \beta I = \begin{pmatrix} 0 & 0 & 1 \\ 2 & 1 - \beta & -2 \\ 3 & 1 & -2 - \beta \end{pmatrix}$$

$$|A - \beta I| = 2 - 3(1 - \beta) = 3\beta - 1 = 0 \Rightarrow \beta = \frac{1}{3}$$

5. Ans (8)



$$S_1P - SP = 20$$

$$\beta - \frac{\delta}{\sin \frac{\alpha}{2}} = 20$$

$$\beta^2 + \frac{\delta^2}{\sin^2 \frac{\alpha}{2}} - 400 = \frac{2\beta\delta}{\sin \frac{\alpha}{2}}$$

$$\frac{1}{\text{SP}} = \frac{\sin \frac{\alpha}{2}}{\delta}$$

$$\cos \alpha = \frac{SP^2 + \beta^2 - 656}{2\beta \frac{\delta}{\sin \frac{\alpha}{2}}}$$

$$= \frac{\frac{2\beta\delta}{\sin \frac{\alpha}{2}} - 256}{\frac{2\beta\delta}{\sin \frac{\alpha}{2}}} = \cos \alpha$$

$$\frac{\lambda - 128}{\lambda} = \cos \alpha$$

$$\lambda(1 - \cos\alpha) = 128$$

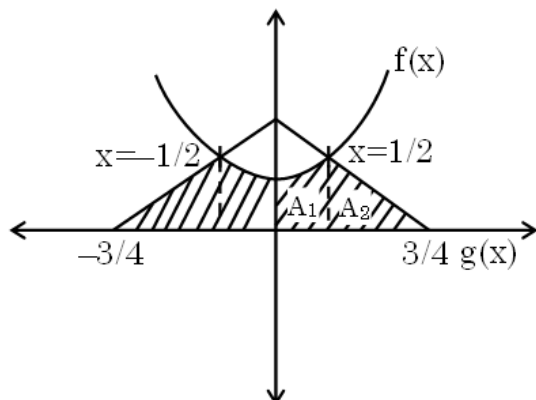
$$\frac{\beta\delta}{\sin \frac{\alpha}{2}}.2\sin^2 \frac{\alpha}{2} = 128$$

$$\beta\delta \sin \frac{\alpha}{2} = 64$$

6. Ans (7)

$$x^2 + \frac{5}{12} = 2 - \frac{8x}{3}$$

$$x^2 + \frac{8x}{3} + \frac{5}{12} - 2 = 0$$



$$12x^2 + 32x - 19 = 0$$

$$12x^2 + 38x - 6x - 19 = 0$$

$$2x(6x + 19) - 1(6x + 19) = 0$$

$$(6x + 19)(2x - 1) = 0$$

$$x = \frac{1}{2}$$

$$\alpha = 3(A_1 + A_2)$$

$$\alpha = 2 \left(\int_0^{1/2} x^2 + \frac{5}{12} dx + \frac{1}{2} \times \frac{1}{4} \times \frac{2}{3} \right)$$

$$\Rightarrow \alpha = 2 \left[\left(\frac{x^3}{3} + \frac{5x}{12} \right)_0^{1/2} + \frac{1}{12} \right]$$

$$\Rightarrow \alpha = 2 \left[\frac{1}{24} + \frac{5}{24} + \frac{1}{12} \right]$$

$$\Rightarrow \alpha = 2 \left[\frac{1+5+2}{24} \right]$$

$$\Rightarrow \alpha = \frac{2}{3}$$

7. Ans (4)

$$\text{Given, } \bar{z} - z^2 = i(\bar{z} + z^2)$$

$$\Rightarrow (1-i)\bar{z} = (1+i)z^2$$

$$\Rightarrow \frac{(1-i)}{(1+i)}\bar{z} = z^2$$

$$\Rightarrow \left(-\frac{2i}{2} \right) \bar{z} = z^2$$

$$\therefore z^2 = -i\bar{z}$$

$$\text{Let } z = x + iy,$$

$$\therefore (x^2 - y^2) + i(2xy) = -i(x - iy)$$

$$\text{so, } x^2 - y^2 + y = 0 \quad \dots(1)$$

$$\text{and } (2y+1)x = 0 \quad \dots(2)$$

$$\Rightarrow x = 0 \text{ or } y = -\frac{1}{2}$$

$$\text{Case I : When } x = 0$$

$$\therefore (1) \Rightarrow y(1-y) = 0 \Rightarrow y = 0, 1$$

$$\therefore (0,0), (0,1)$$

$$\text{Case II : When } y = -\frac{1}{2}$$

$$\therefore (1) \Rightarrow x^2 - \frac{1}{4} - \frac{1}{2} = 0 \Rightarrow x^2 = \frac{3}{4}$$

$$\Rightarrow x = \pm \frac{\sqrt{3}}{2}$$

$$\therefore \left(\frac{\sqrt{3}}{2}, -\frac{1}{2} \right), \left(-\frac{\sqrt{3}}{2}, -\frac{1}{2} \right)$$

$$\Rightarrow \text{Number of distinct 'z' is equal to 4.}$$

8. Ans (9)

Given

$$A_{51} - A_{50} = 1000 \Rightarrow \ell_{51}w_{51} - \ell_{50}w_{50} = 1000$$

$$\Rightarrow (\ell_1 + 50d_1)(w_1 + 50d_2) - (\ell_1 + 49d_1)(w_1 + 49d_2) = 1000$$

$$\Rightarrow (\ell_1 d_2 + w_1 d_1) = 10 \quad \dots(1)$$

$$(\text{As } d_1 d_2 = 10)$$

$$\therefore A_{100} - A_{90} = \ell_{100}w_{100} - \ell_{90}w_{90}$$

$$= (\ell_1 + 99d_1)(w_1 + 99d_2) - (\ell_1 + 89d_1)(w_1 + 89d_2)$$

$$= 10(\ell_1 d_2 + w_1 d_1) + (99^2 - 89^2) d_1 d_2$$

$$= 10(10) + \underbrace{(99-89)}_{=10} (99+89)(10)$$

$$(\text{As, } d_1 d_2 = 10)$$

$$= 100(1 + 188) = 100(189)$$

$$= 18900$$